



Université de Technologie D'Haïti (UNITECH)

FACULTE DES SCIENCES INFORMATIQUES

Niveau IV

Devoir

Du cours

Intro a la Cyber sécurité

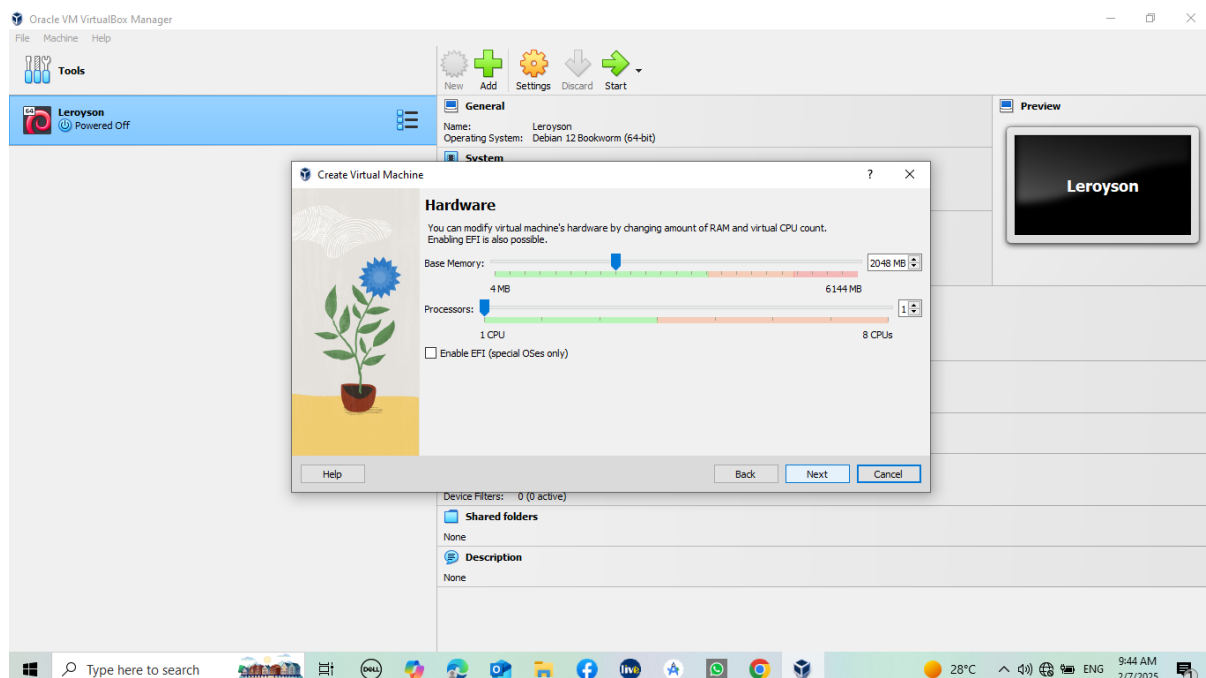
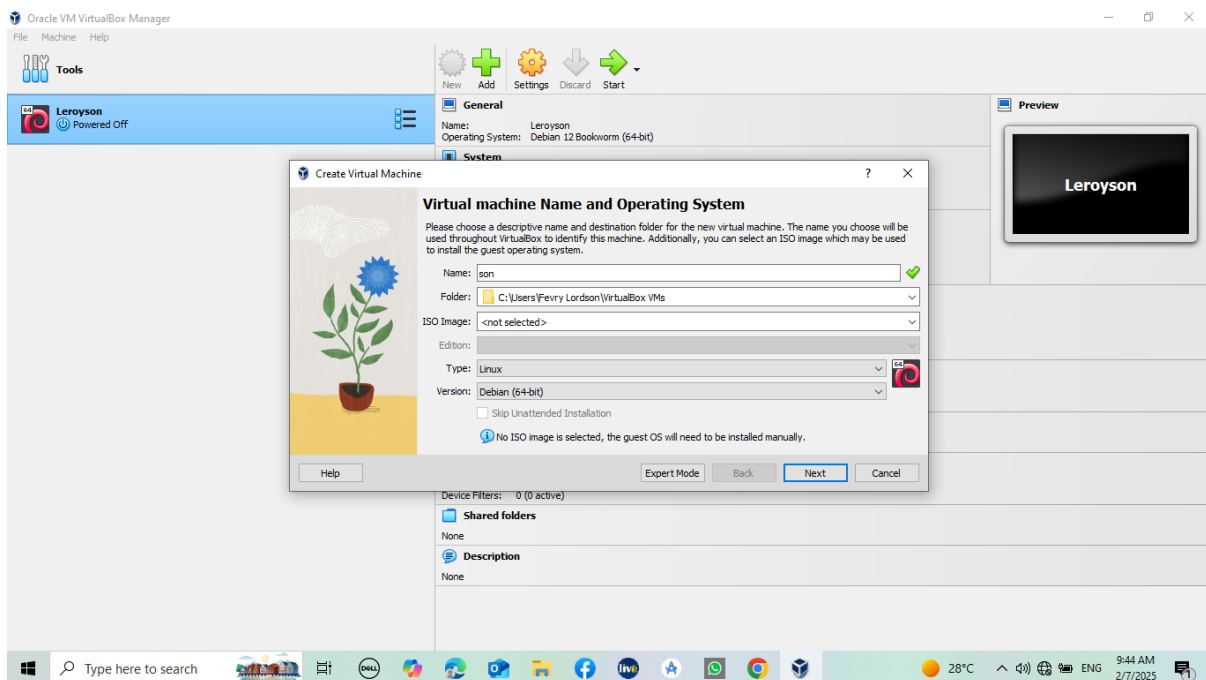
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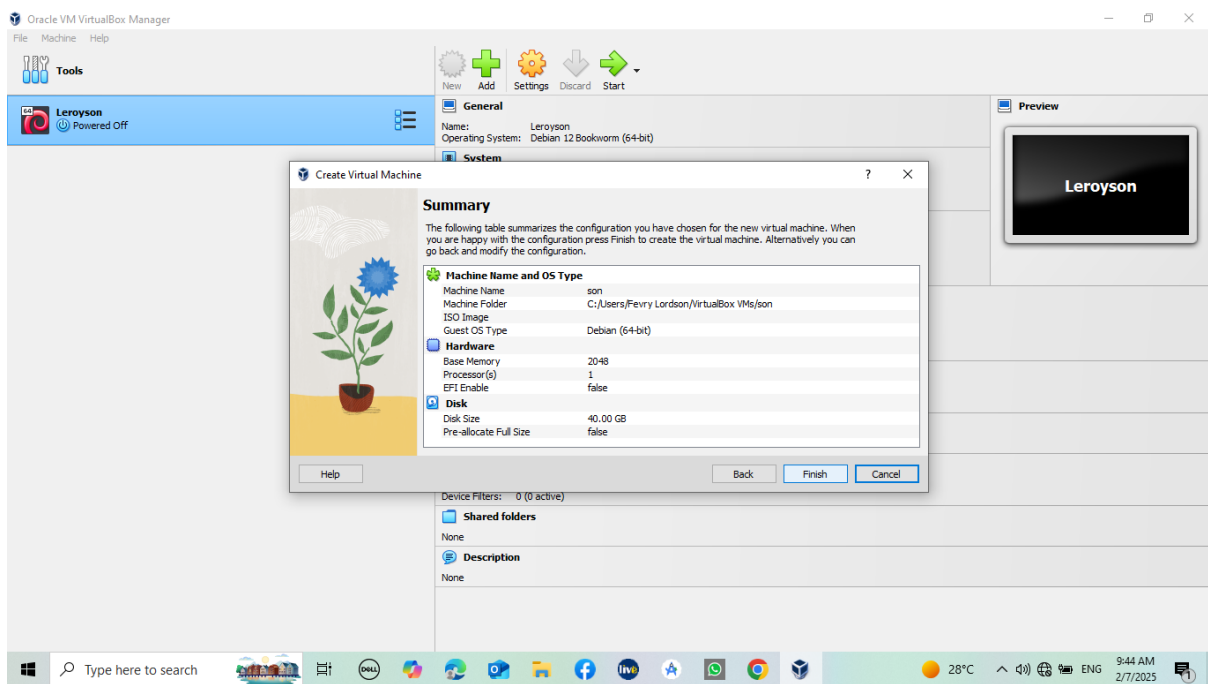
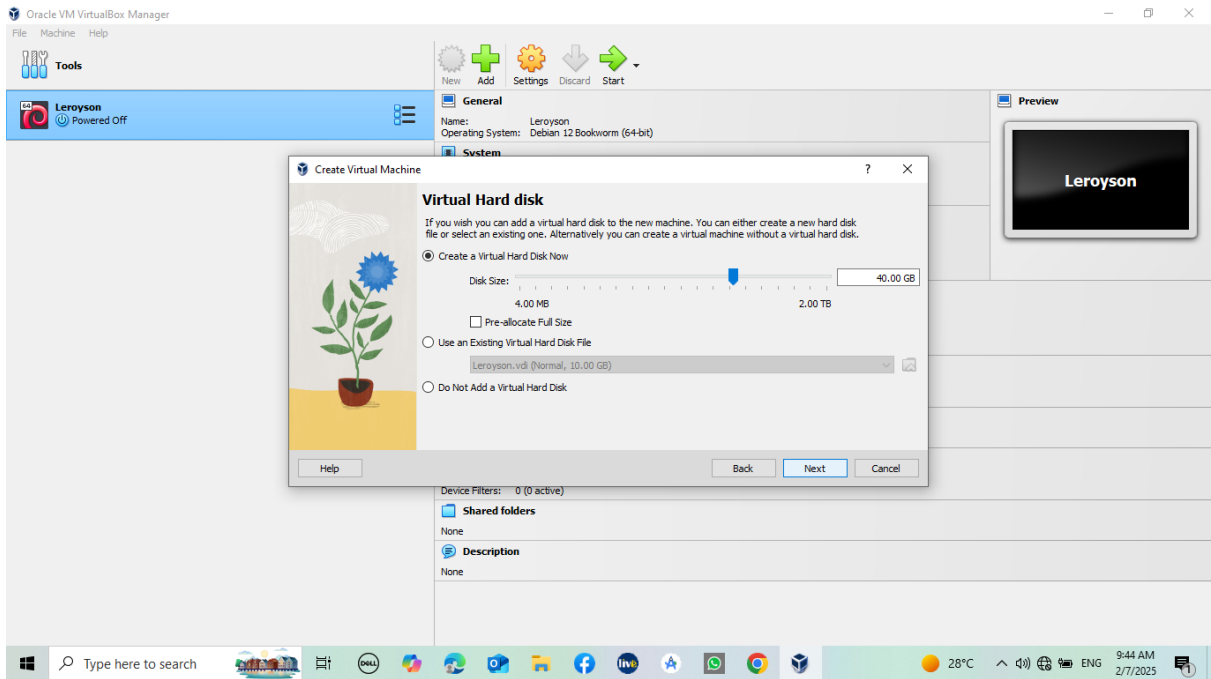
Lordson FEVRY

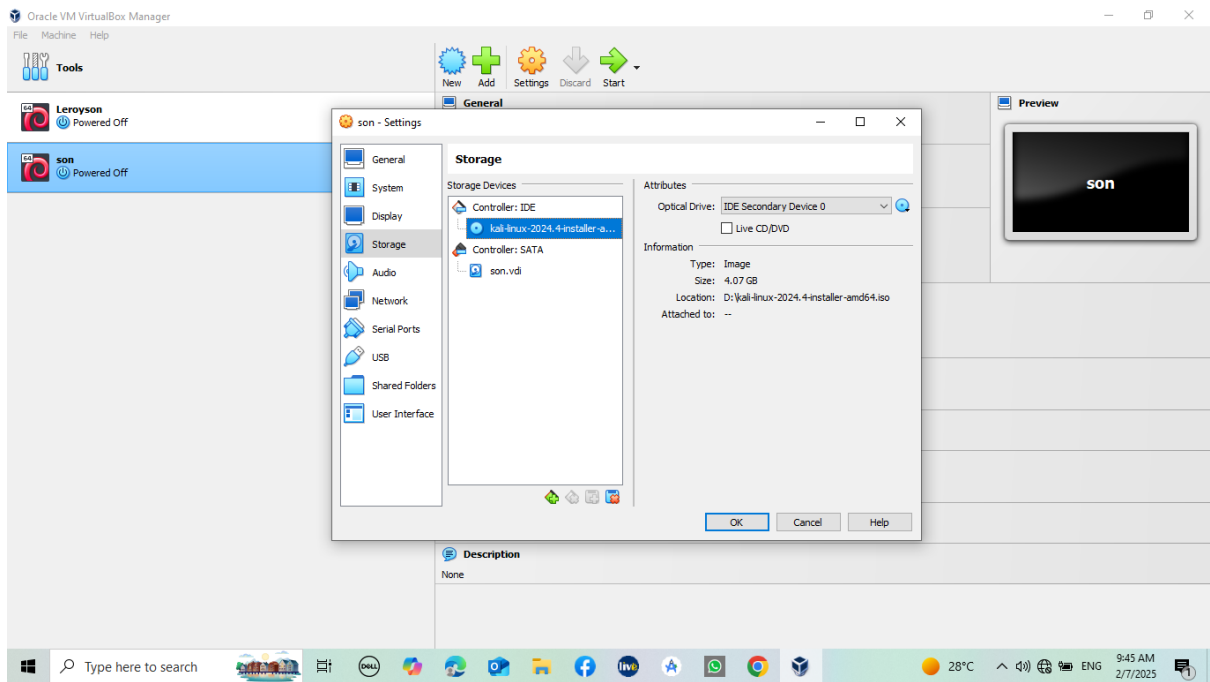
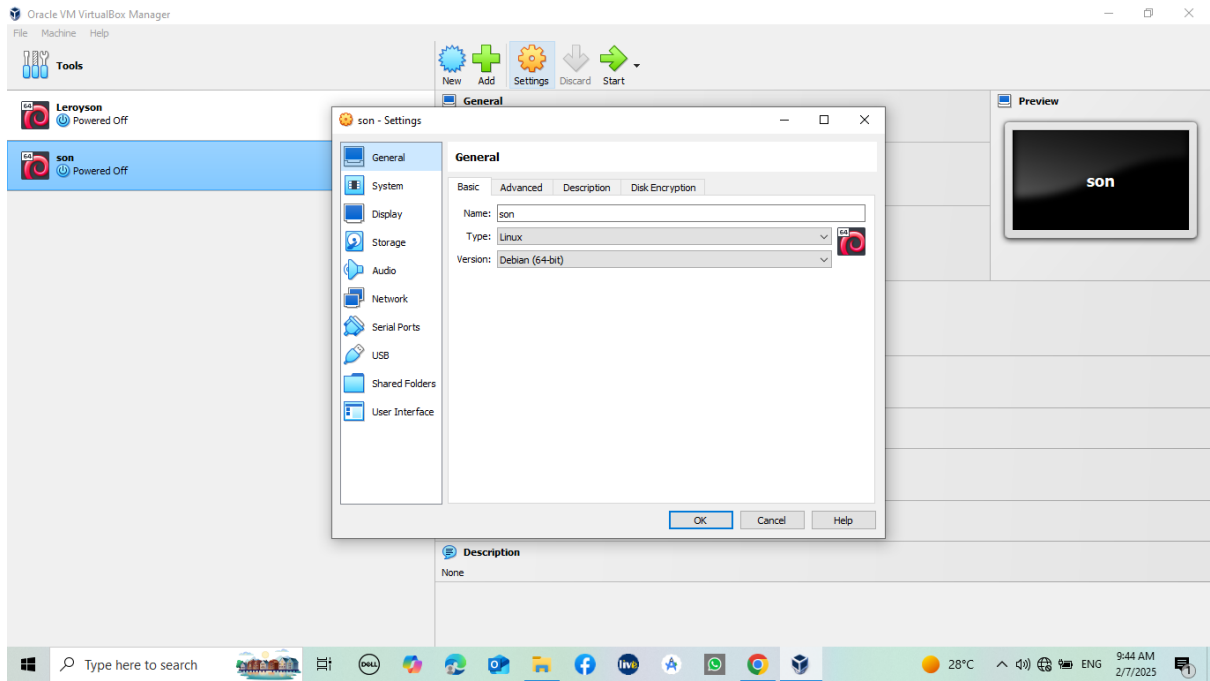
Soumis au Professeur Mr **Ismael Saint Amour**

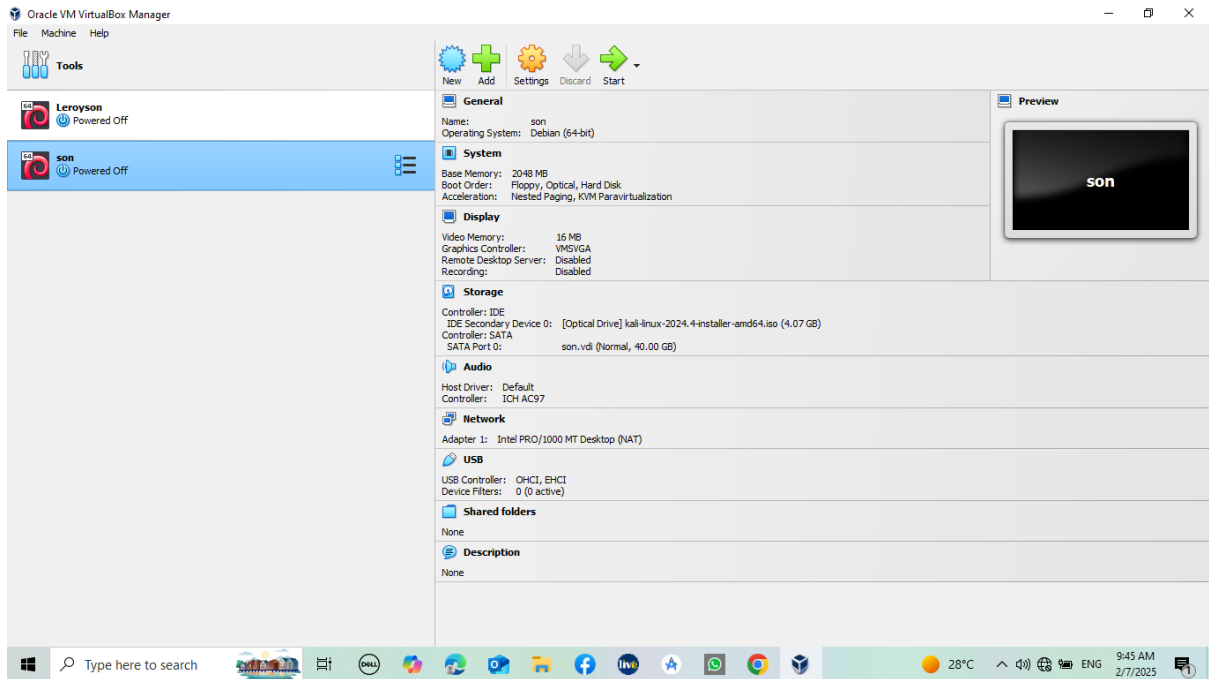
Sujet: TD2 sécurité-informatique.

Création d'une nouvelle machine virtuelle dans VirtualBox

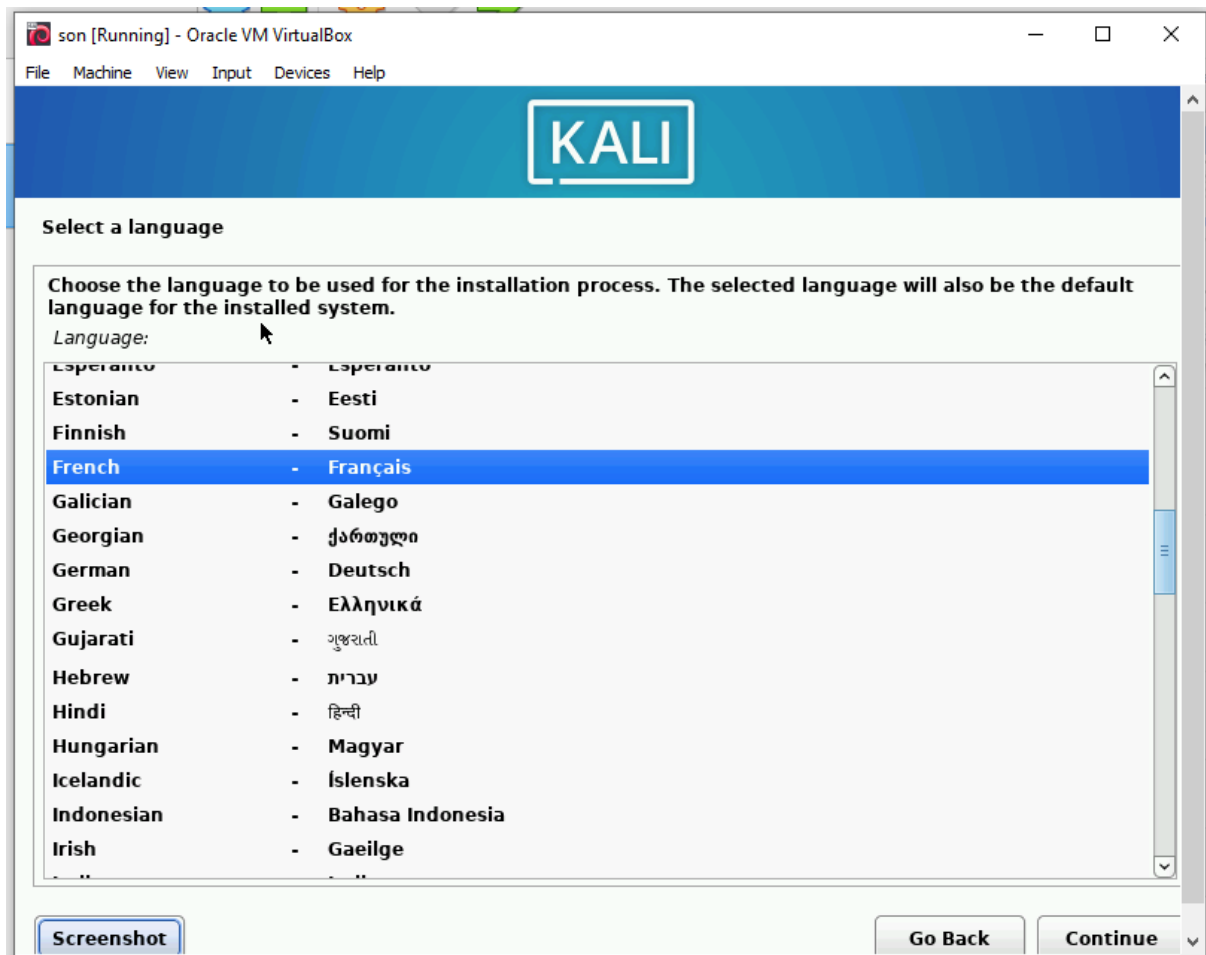


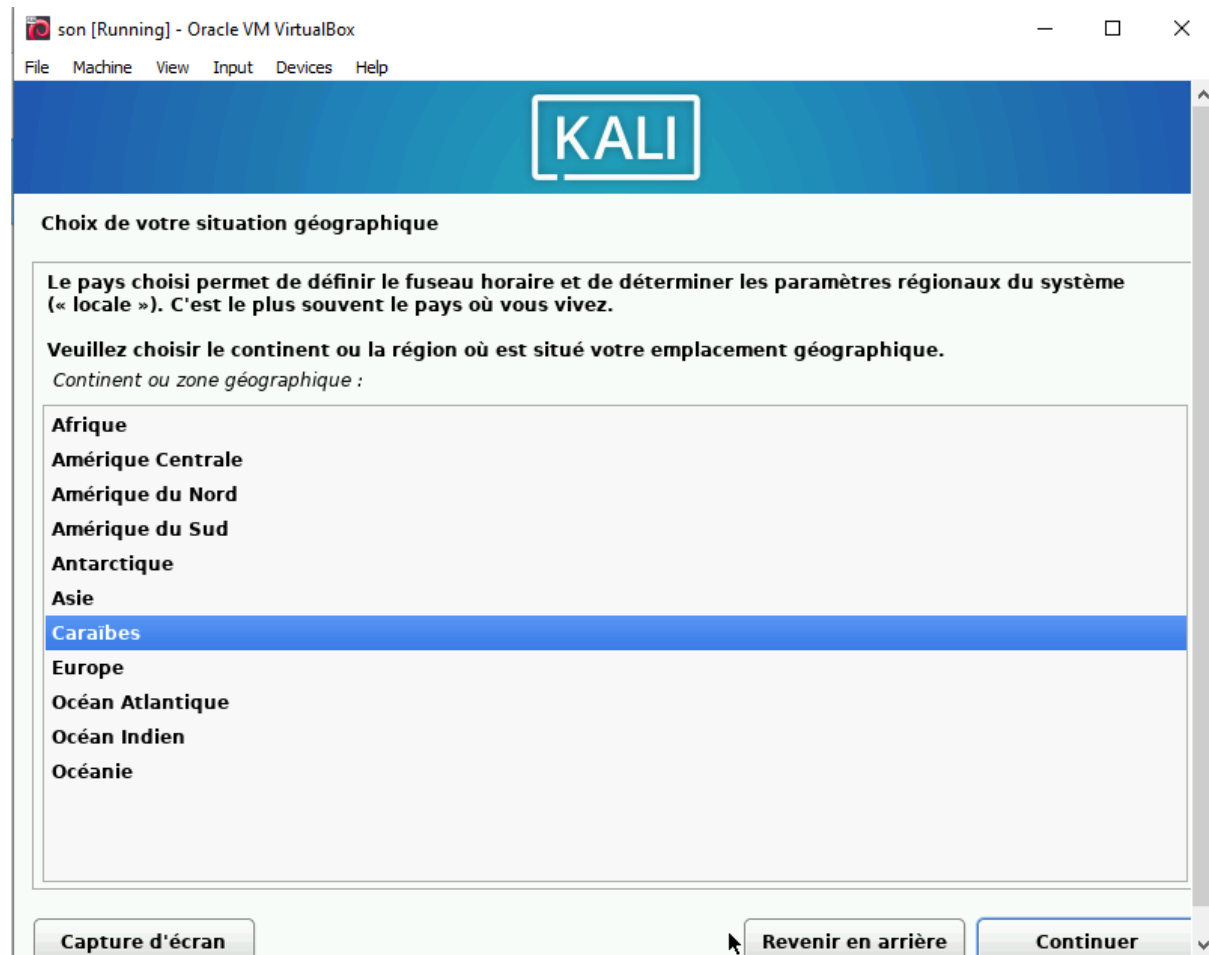
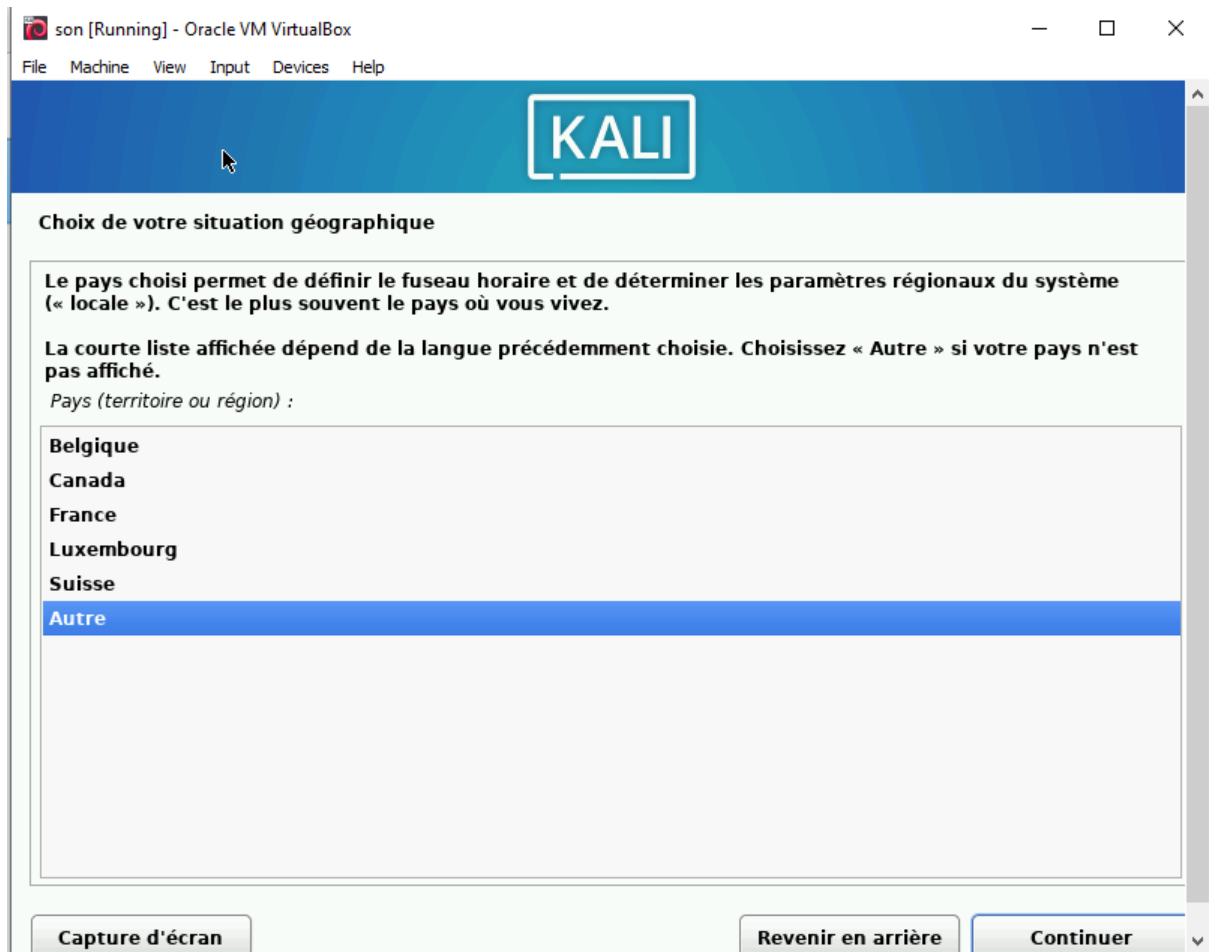


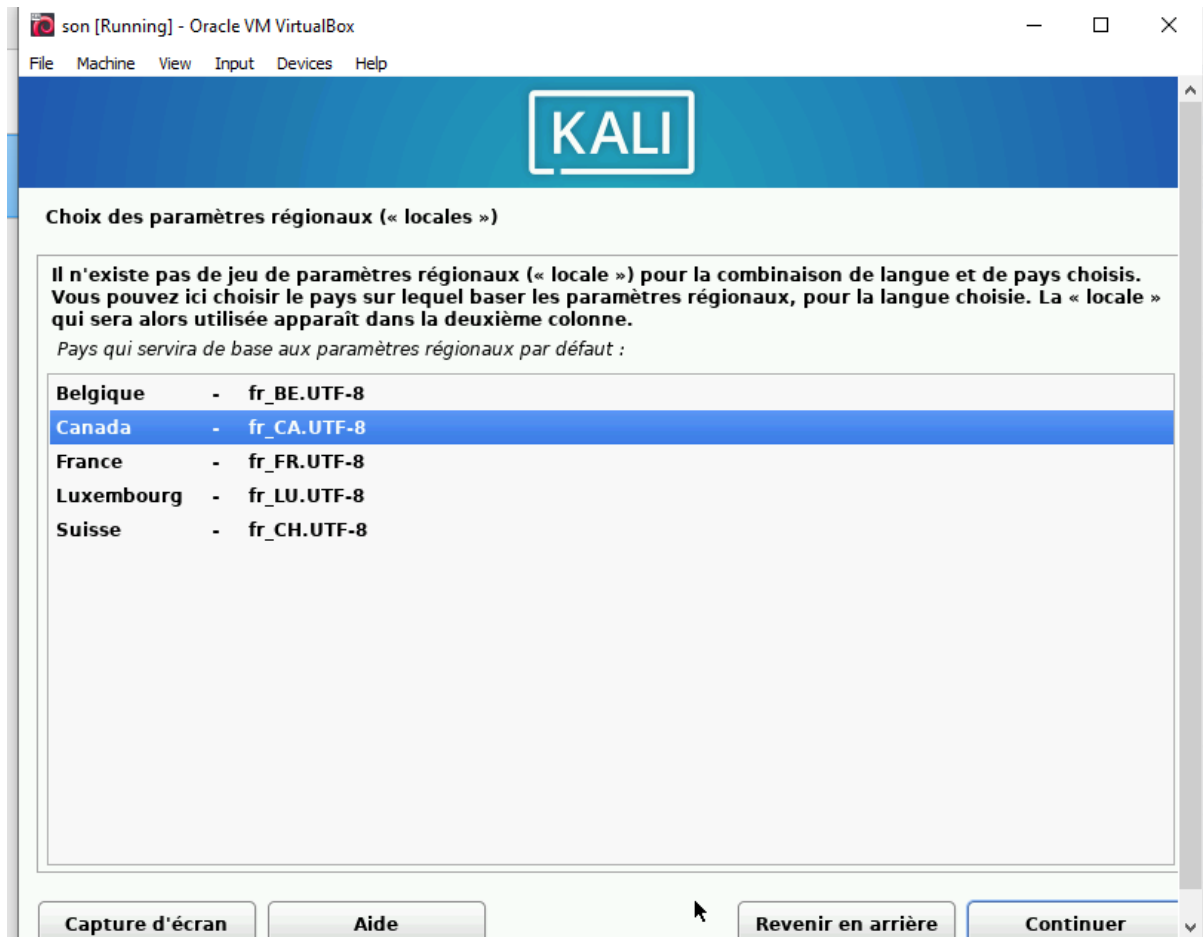
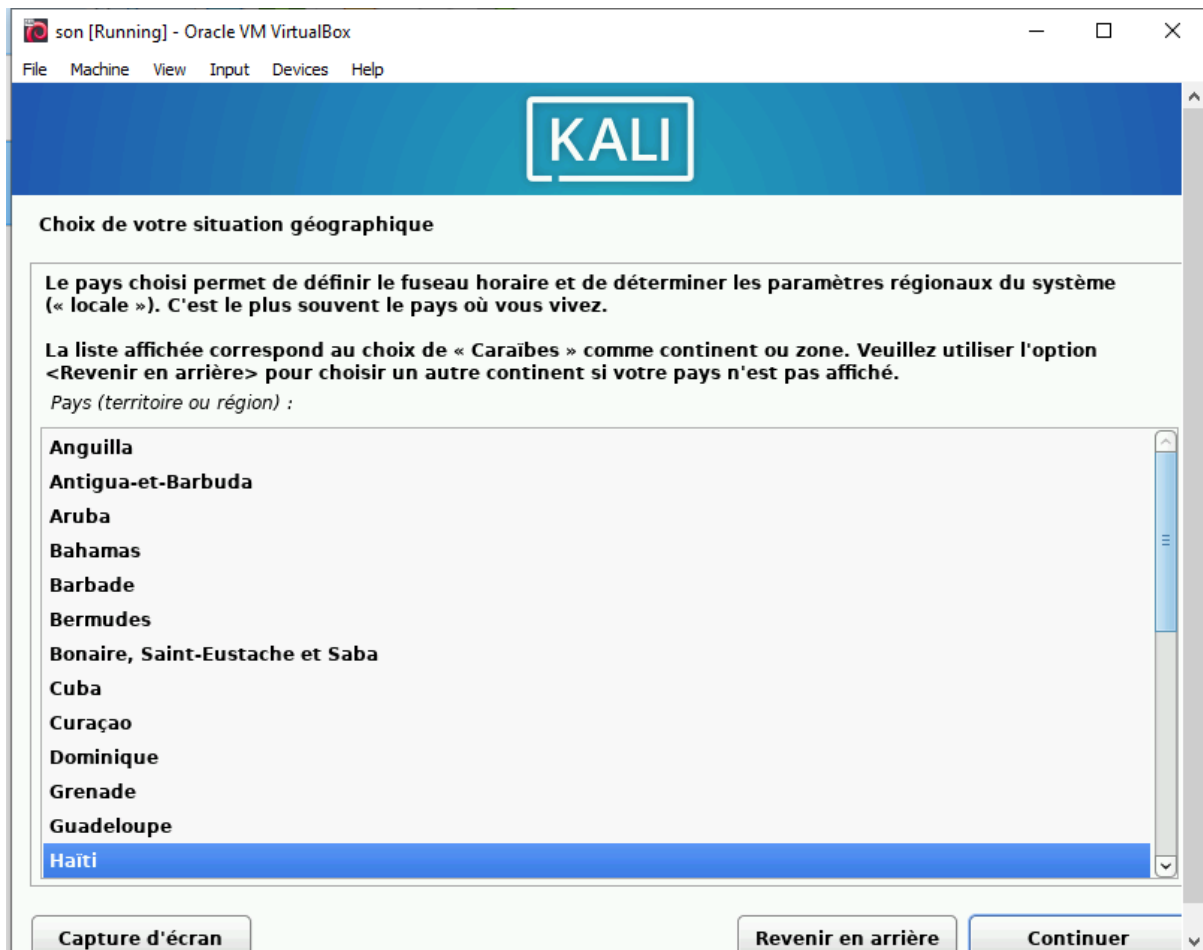


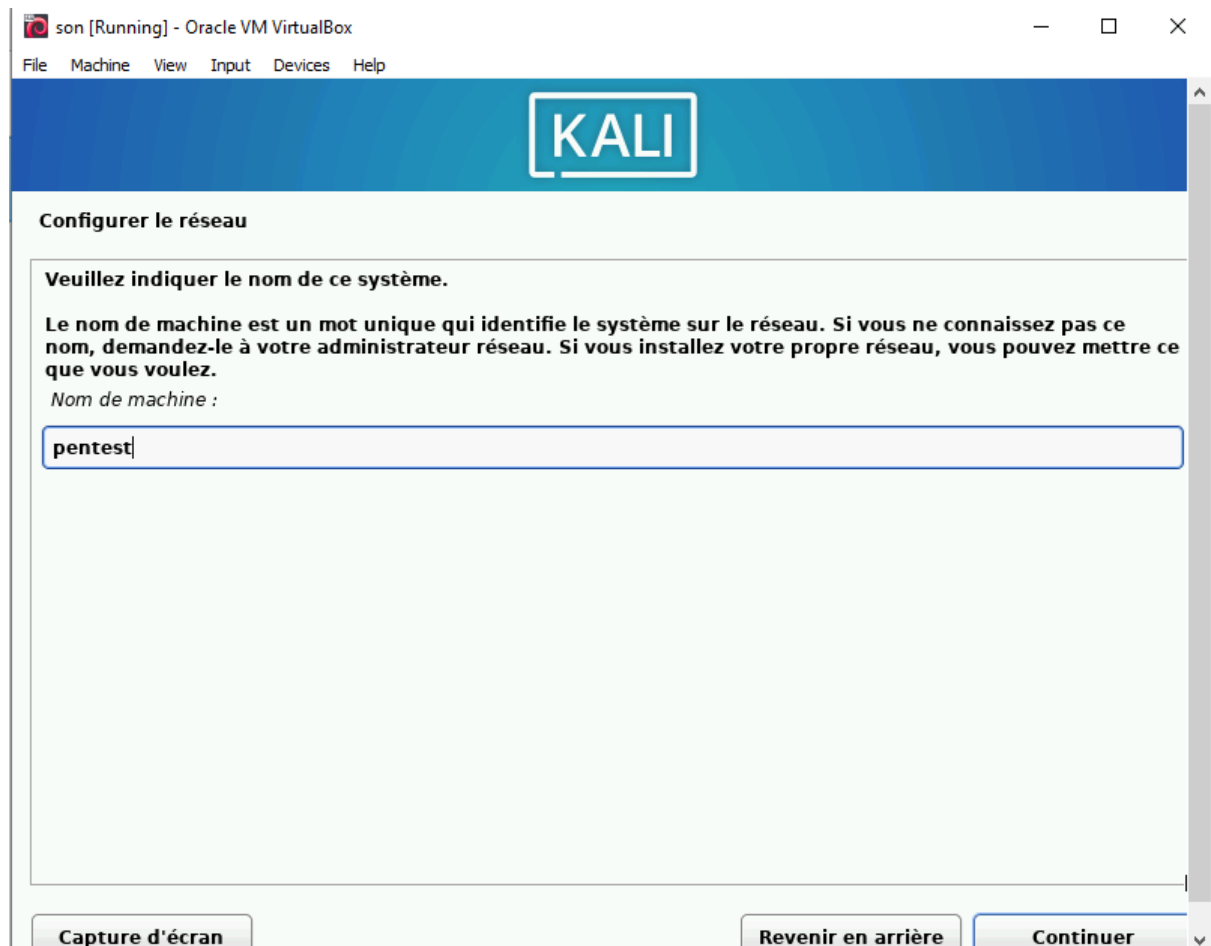
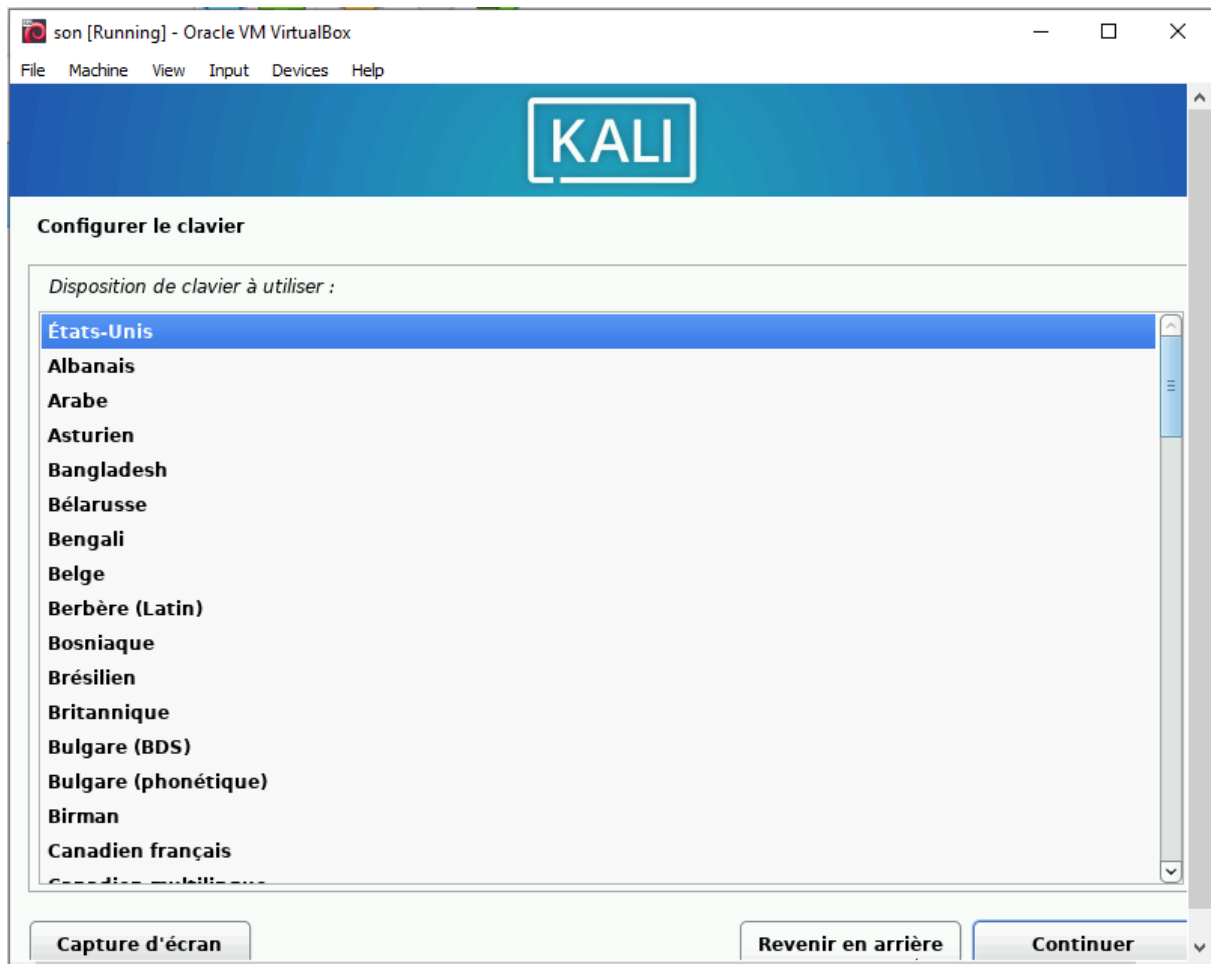


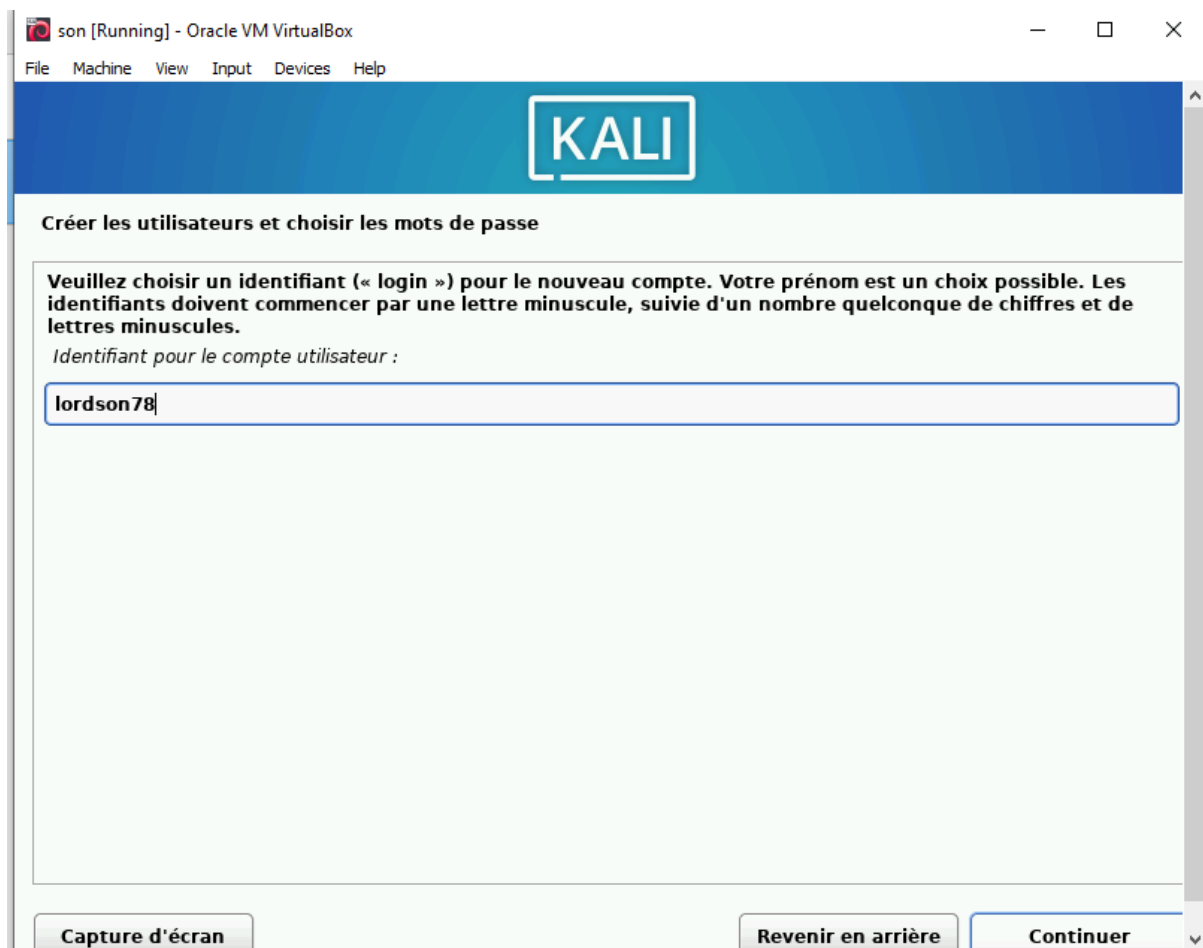
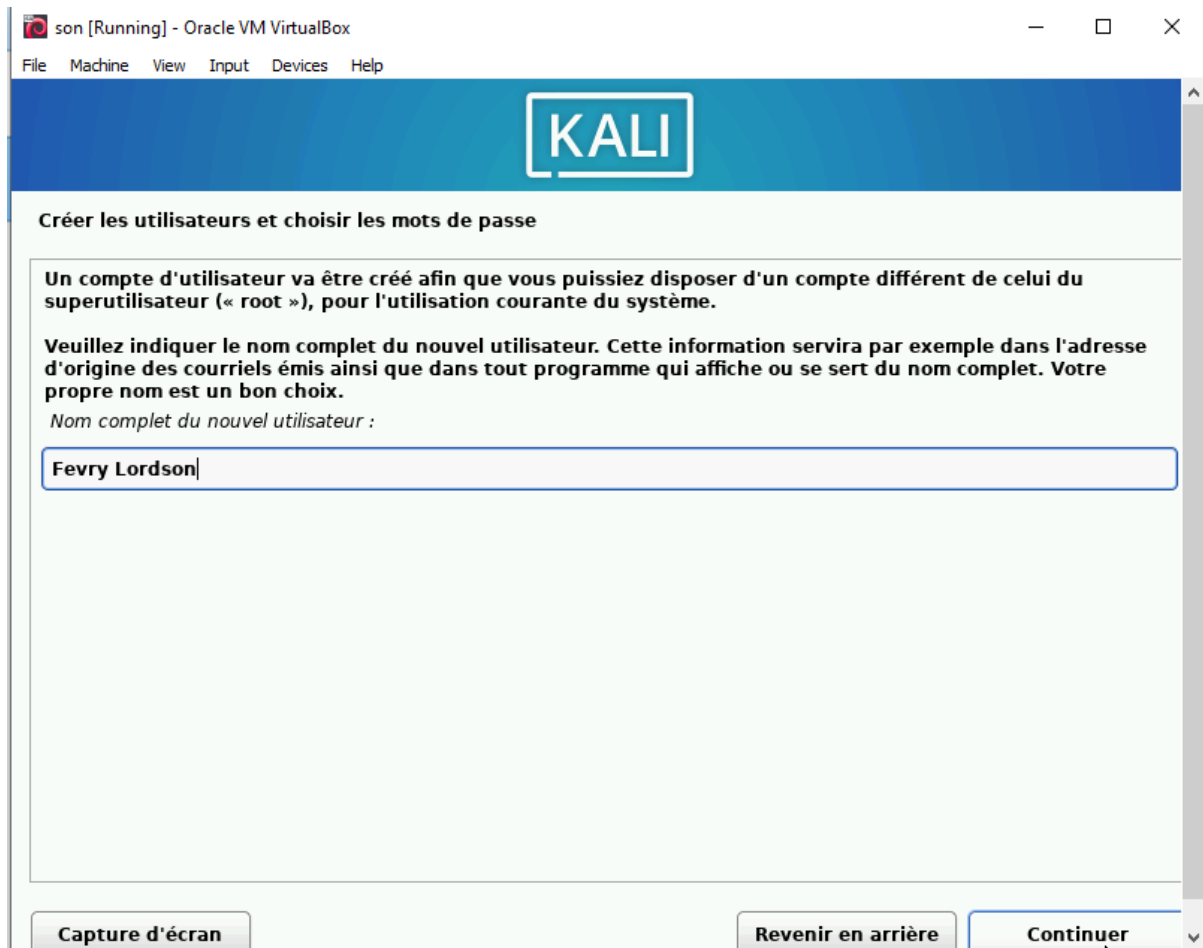
Installation de Kali Linux en suivant les étapes d'installation graphique

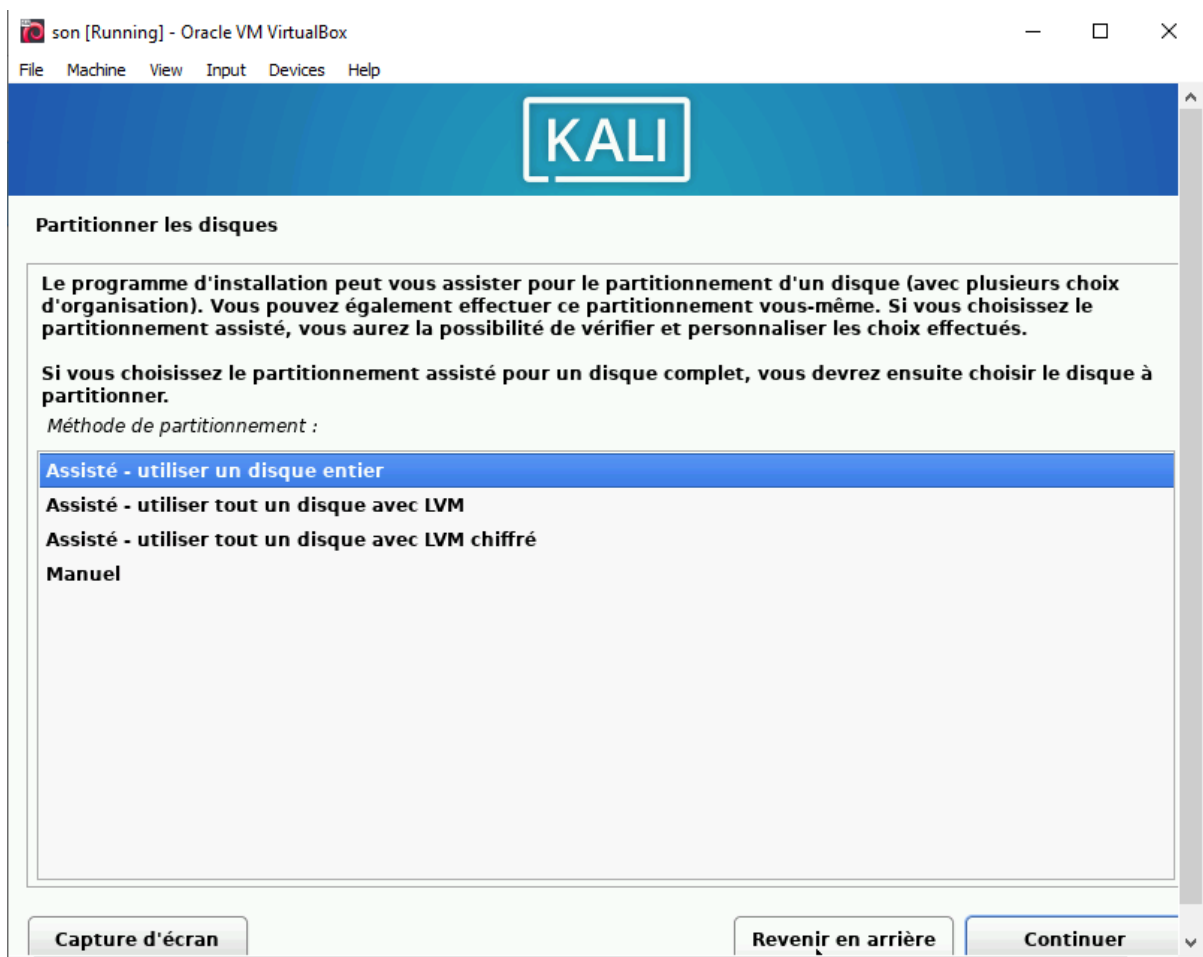
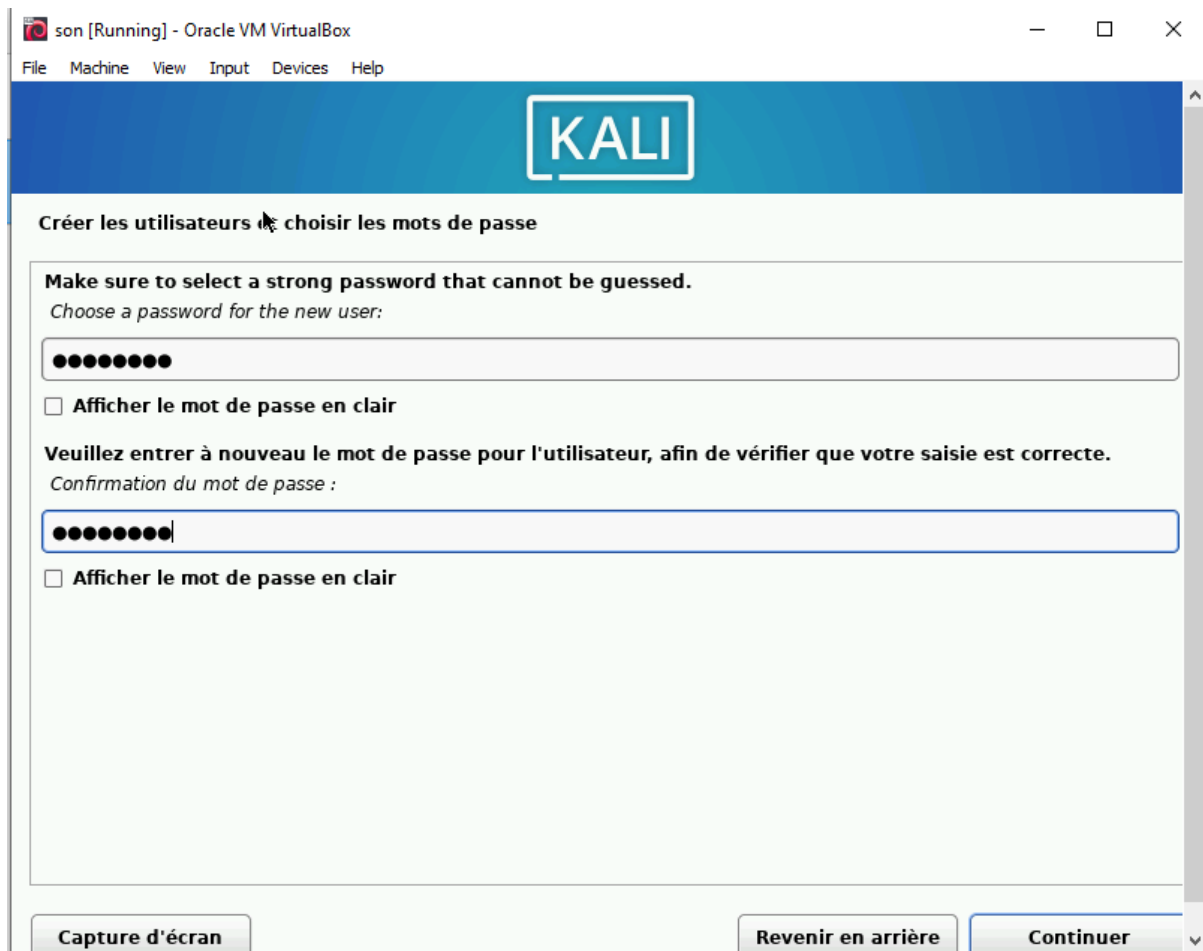


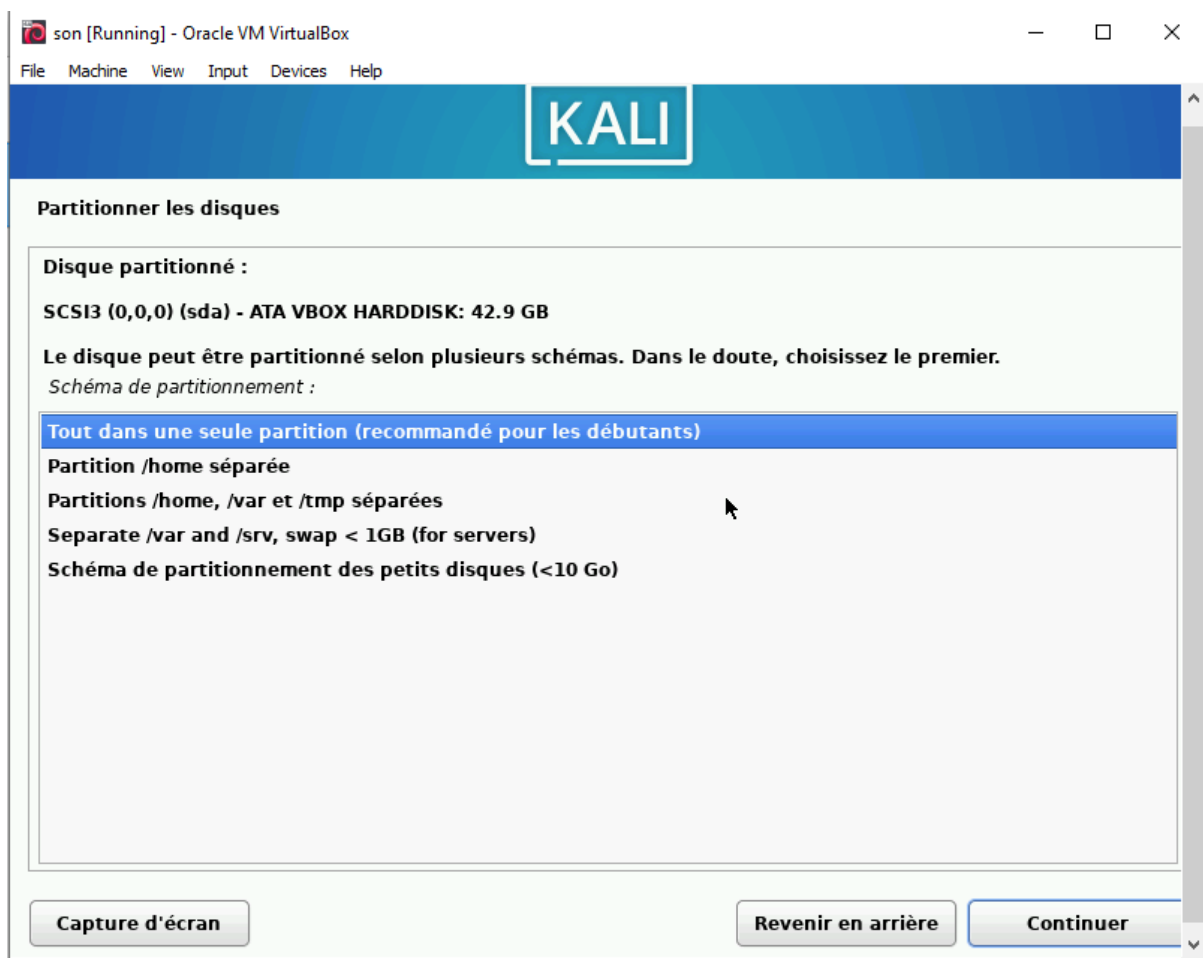
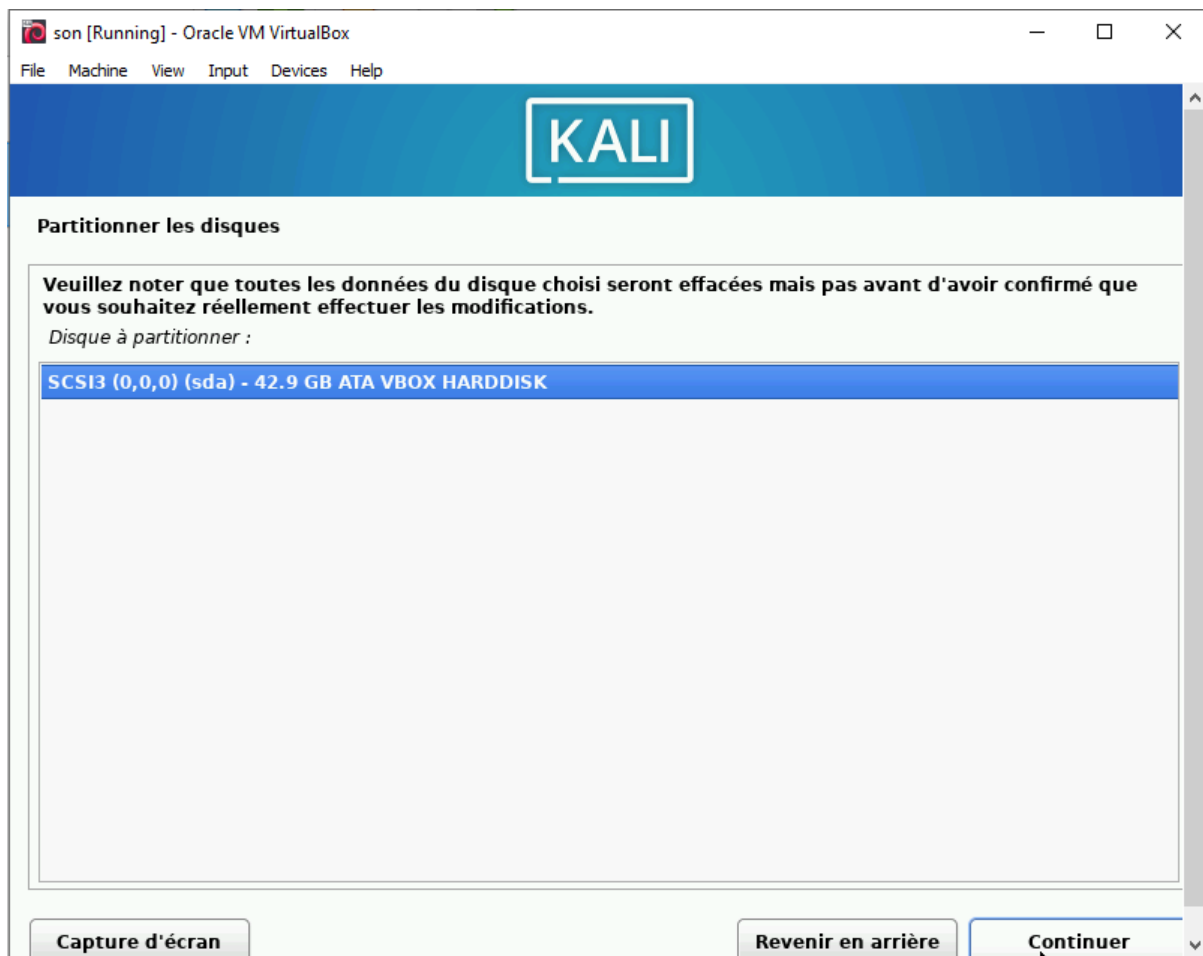


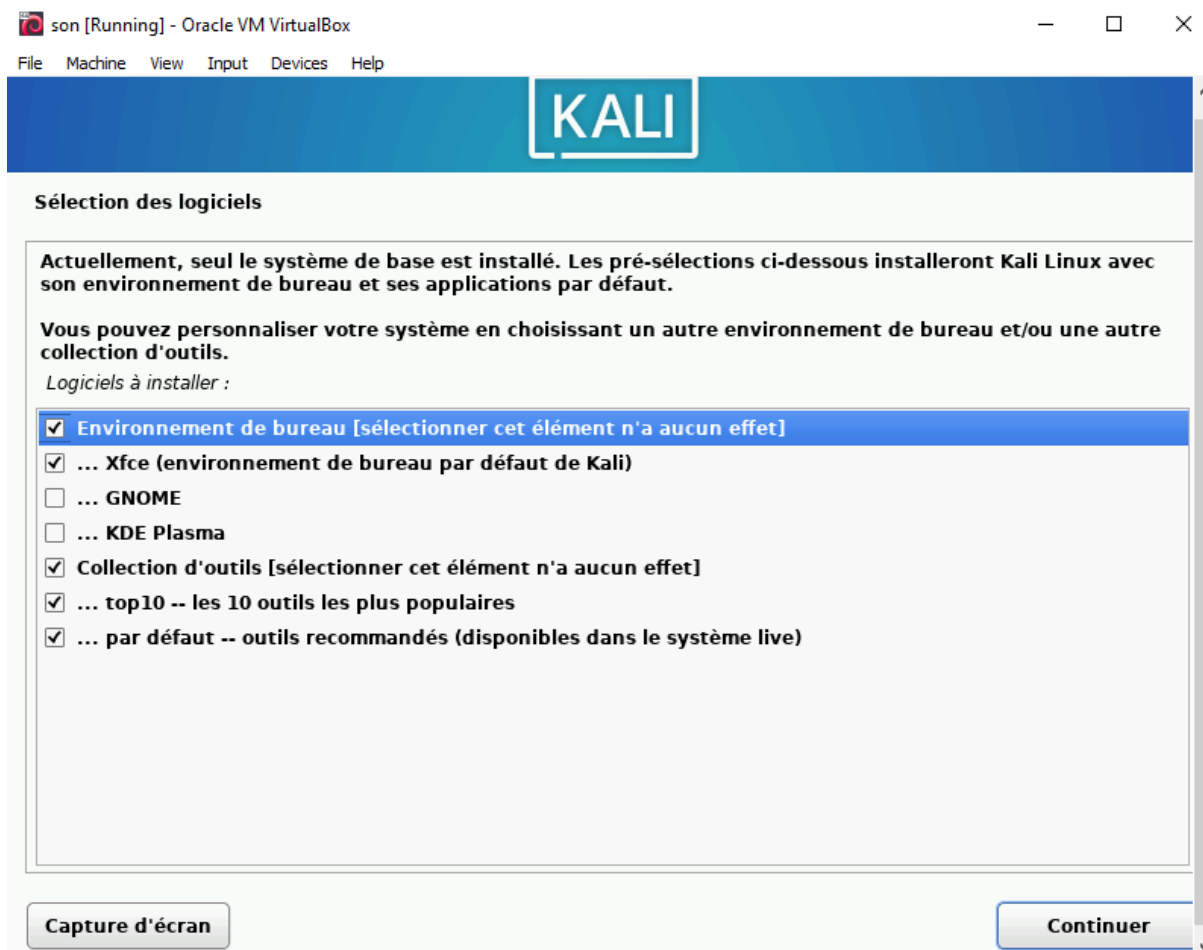
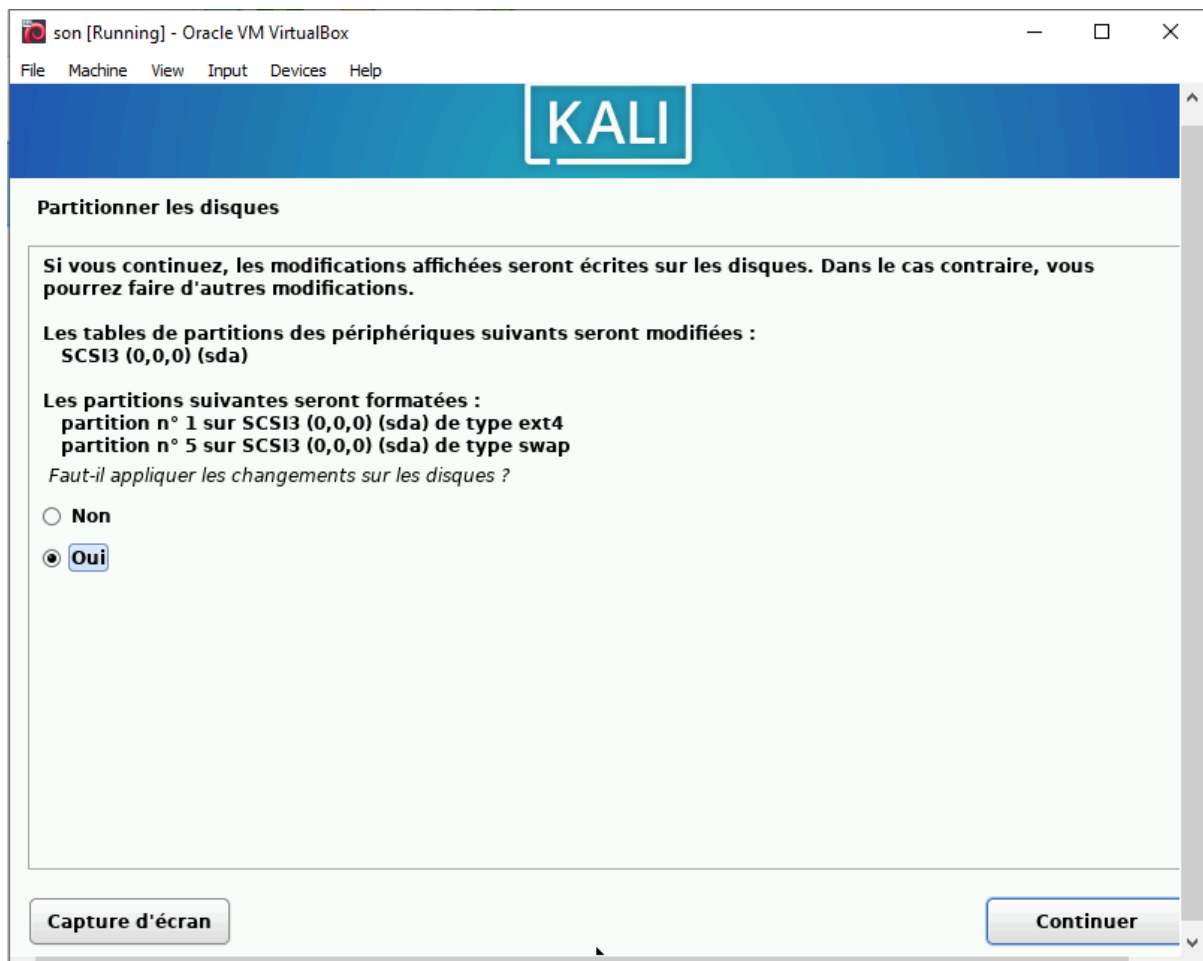


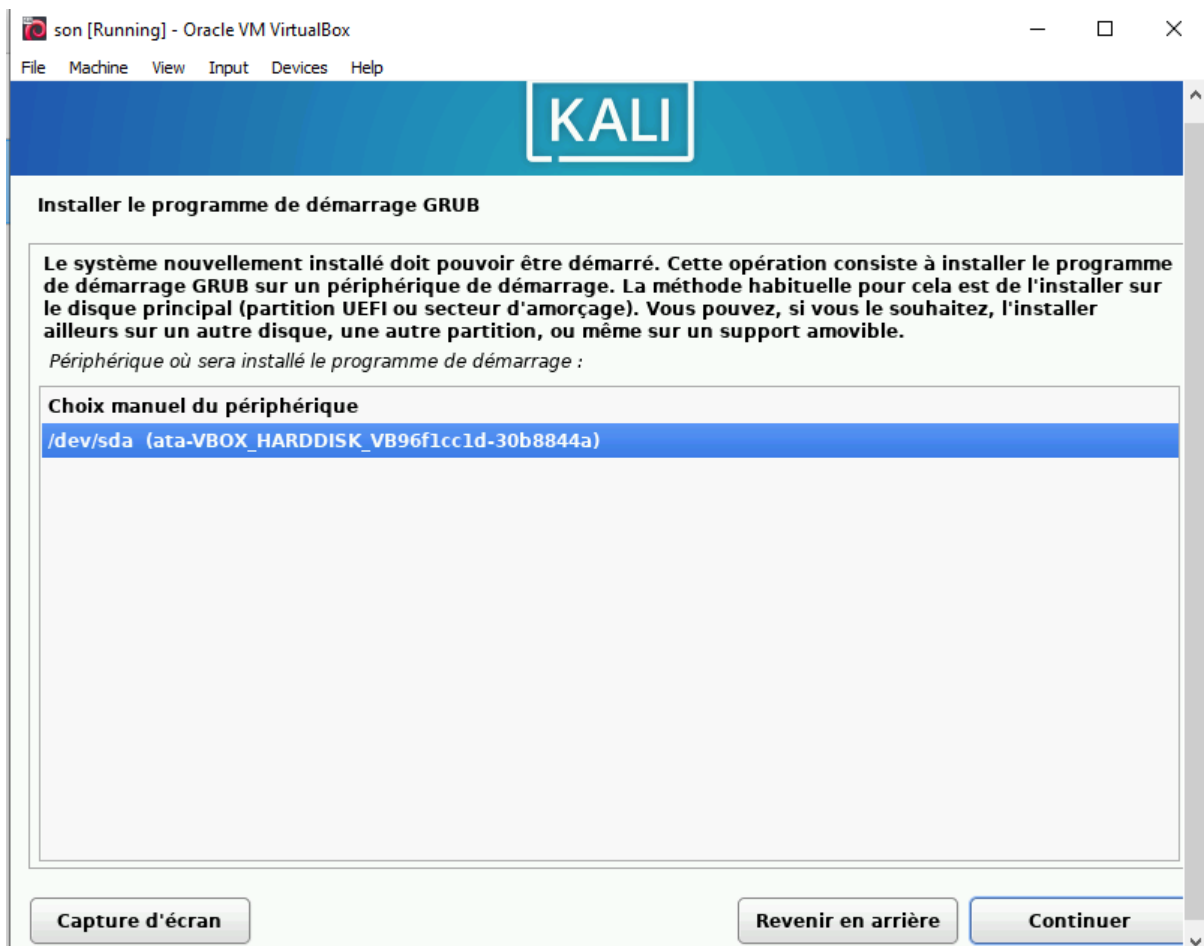
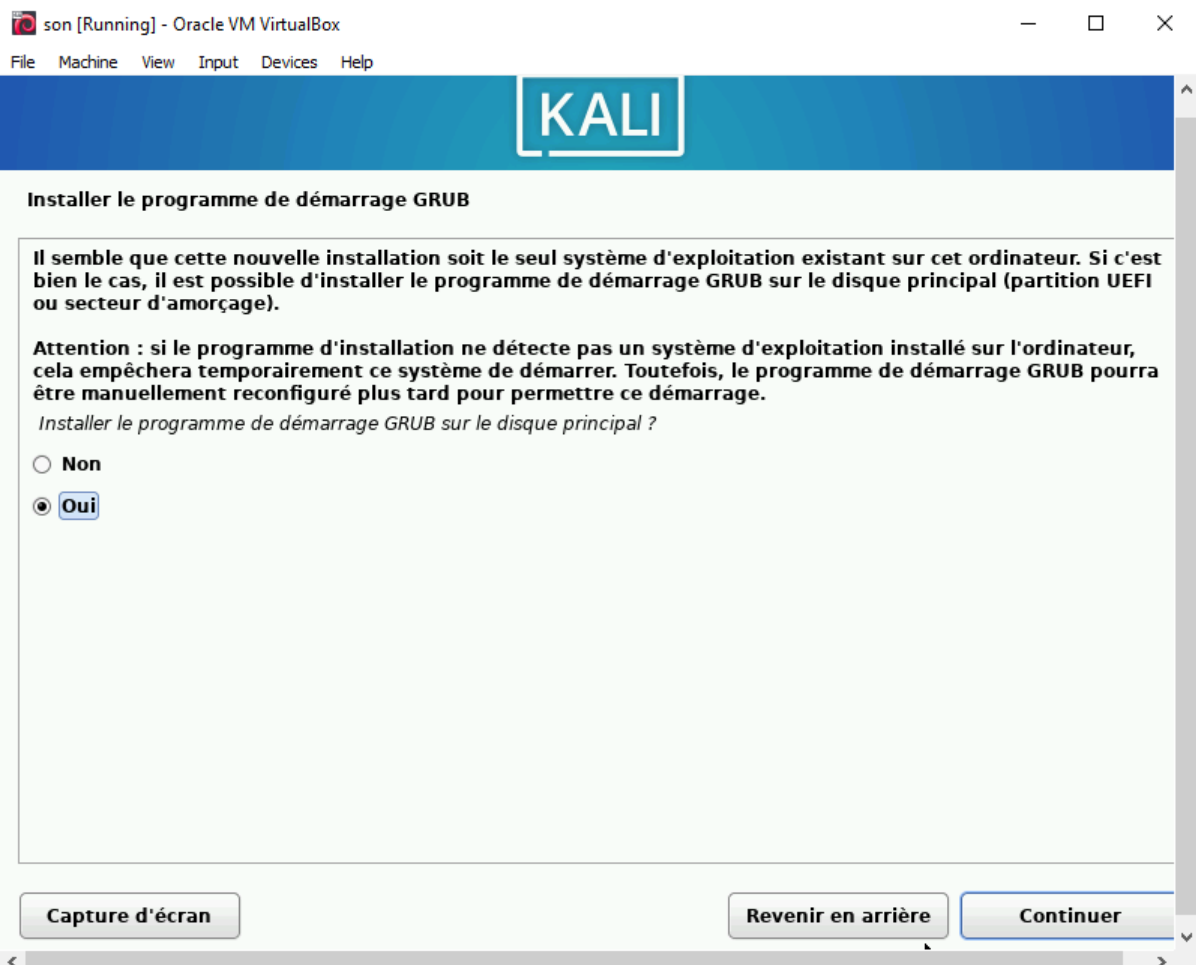


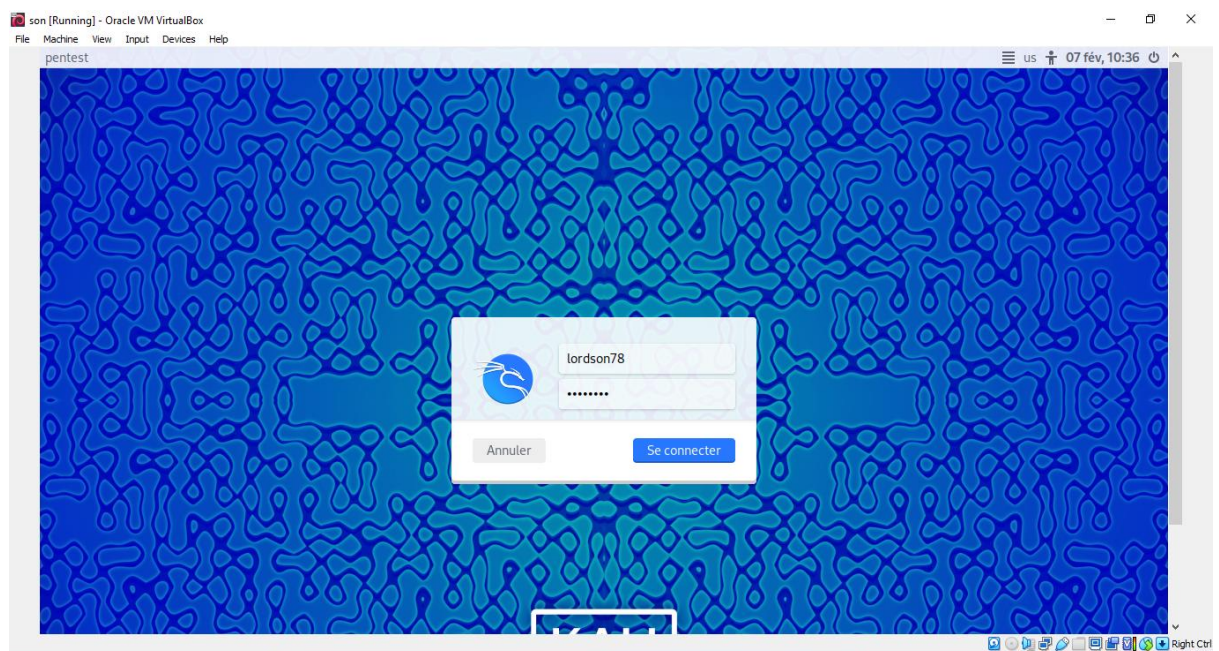
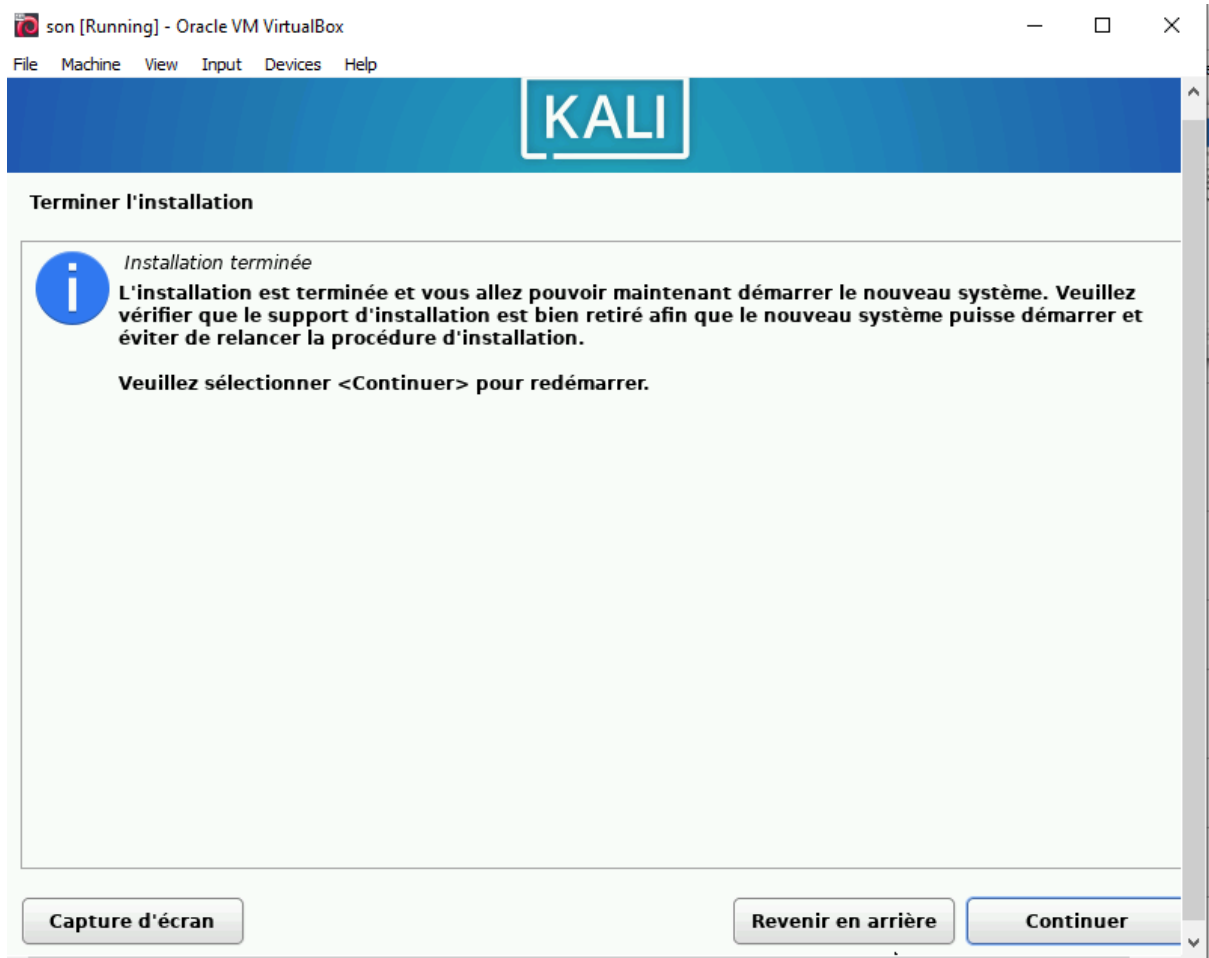












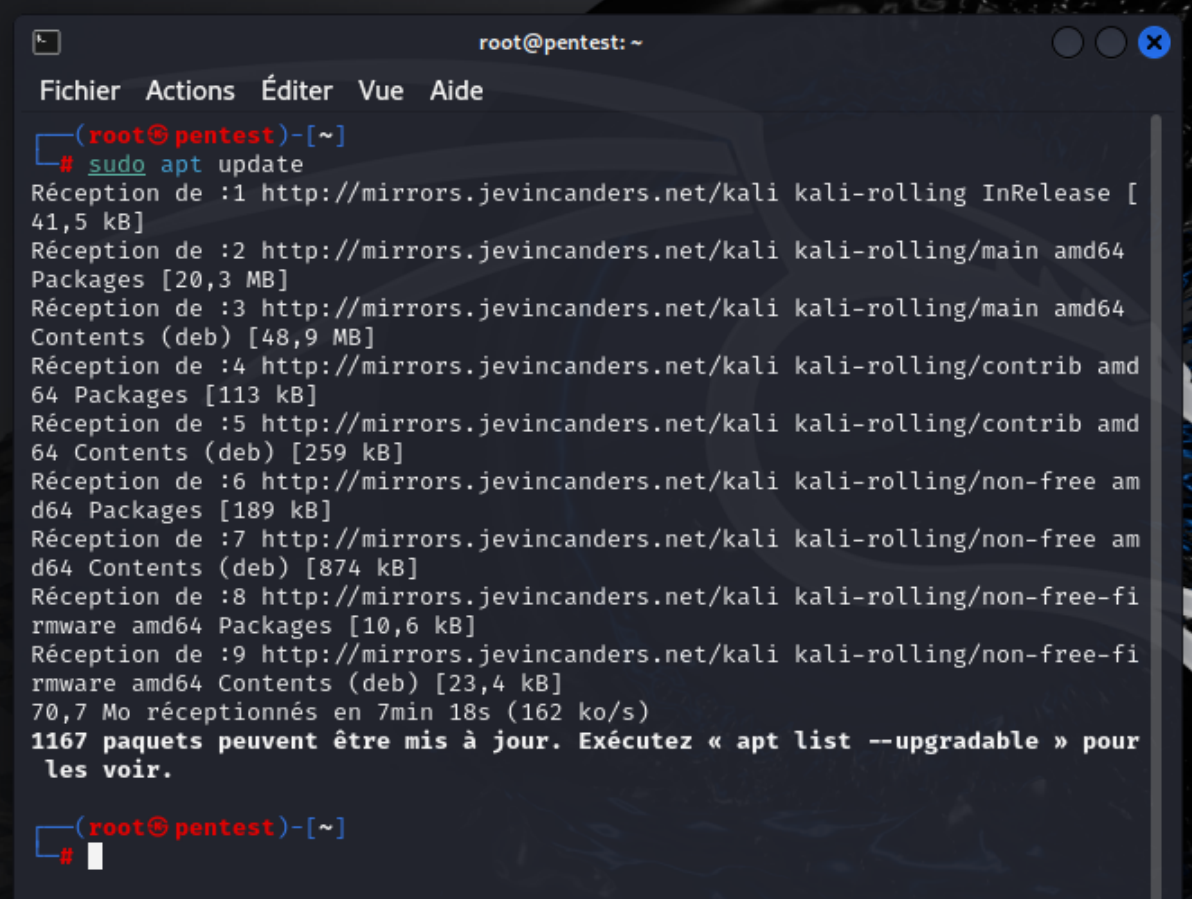


Mise à jour du système après l'installation :

`sudo apt update`

`sudo apt upgrade -y`

`sudo apt dist-upgrade -y`



```
root@pentest: ~
Fichier Actions Éditer Vue Aide
(root@pentest)-[~]
# sudo apt update
Réception de :1 http://mirrors.jevincanders.net/kali kali-rolling InRelease [
41,5 kB]
Réception de :2 http://mirrors.jevincanders.net/kali kali-rolling/main amd64
Packages [20,3 MB]
Réception de :3 http://mirrors.jevincanders.net/kali kali-rolling/main amd64
Contents (deb) [48,9 MB]
Réception de :4 http://mirrors.jevincanders.net/kali kali-rolling/contrib amd
64 Packages [113 kB]
Réception de :5 http://mirrors.jevincanders.net/kali kali-rolling/contrib amd
64 Contents (deb) [259 kB]
Réception de :6 http://mirrors.jevincanders.net/kali kali-rolling/non-free am
d64 Packages [189 kB]
Réception de :7 http://mirrors.jevincanders.net/kali kali-rolling/non-free am
d64 Contents (deb) [874 kB]
Réception de :8 http://mirrors.jevincanders.net/kali kali-rolling/non-free-fi
rmware amd64 Packages [10,6 kB]
Réception de :9 http://mirrors.jevincanders.net/kali kali-rolling/non-free-fi
rmware amd64 Contents (deb) [23,4 kB]
70,7 Mo réceptionnés en 7min 18s (162 ko/s)
1167 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour
les voir.

(root@pentest)-[~]
#
```



```
root@pentest: ~
Fichier Actions Éditer Vue Aide

wireshark
libqt6opengl6:amd64

(root@pentest)-[~]
# sudo apt upgrade -y
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
libbfbio1 libgtksourceviewmm-3.0-0v5
libc++1-19 libhdf5-hl-100t64
libc++abi1-19 libjxl0.9
libcapstone4 libmbedcrypto7t64
libconfig++9v5 libpaper1
libconfig9 libsuperlu6
libdirectfb-1.7-7t64 libtag1v5
libegl-dev libtag1v5-vanilla
libfmt9 libtagc0
libgl1-mesa-dev libunwind-19
libgles-dev libwebrtc-audio-processing1
libgles1 libx265-209
libglvnd-core-dev openjdk-23-jre
libglvnd-dev openjdk-23-jre-headless
libgtksourceview-3.0-1 python3-appdirs
libgtksourceview-3.0-common
Veuillez utiliser « sudo apt autoremove » pour les supprimer.

Upgrading:
kali-linux-default pyqt5-dev-tools
```

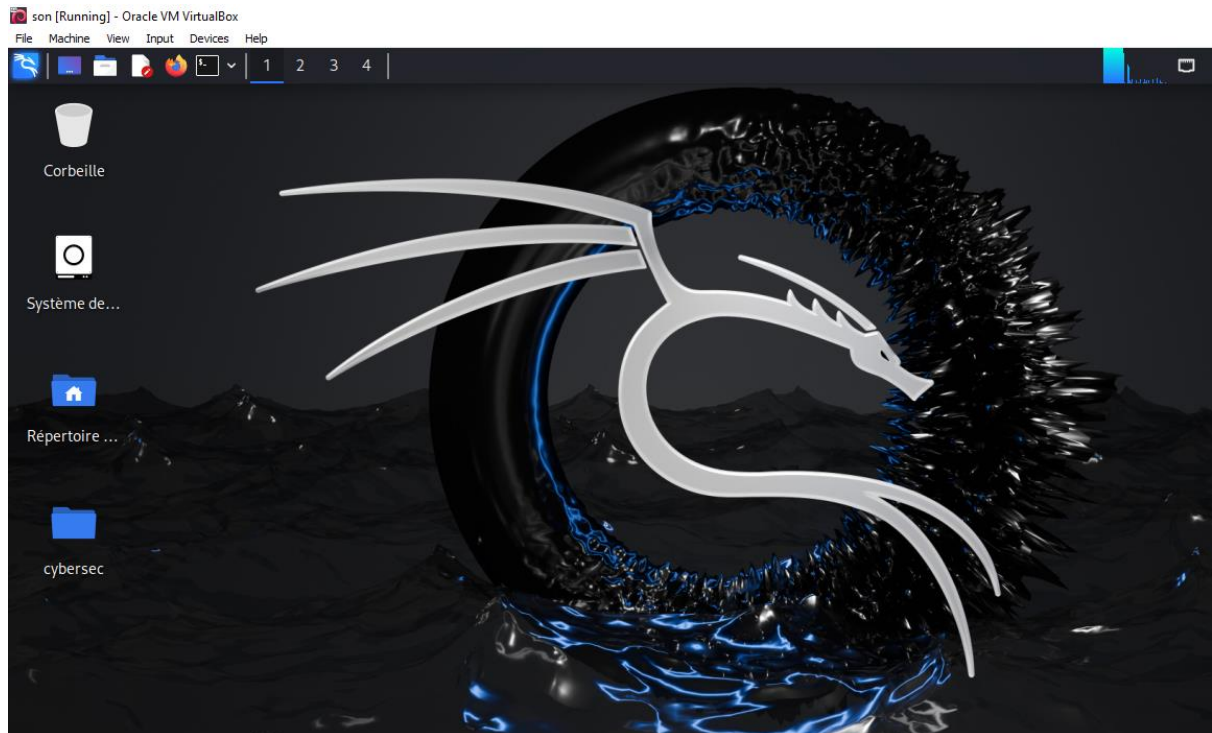
```
root@pentest: ~
Fichier Actions Éditer Vue Aide

(root@pentest)-[~]
# sudo apt dist-upgrade -y
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
imagemagick-6.q16 libhdf5-hl-100t64
libbfbio1 libjxl0.9
libc++1-19 libmagickcore-6.q16-7-extra
libc++abi1-19 libmagickcore-6.q16-7t64
libcapstone4 libmagickwand-6.q16-7t64
libconfig++9v5 libmbedcrypto7t64
libconfig9 libpaper1
libdirectfb-1.7-7t64 libqt5*11extras5
libegl-dev libsuperlu6
libfmt9 libtag1v5
libgl1-mesa-dev libtag1v5-vanilla
libgles-dev libtagc0
libgles1 libunwind-19
libglvnd-core-dev libwebrtc-audio-processing1
libglvnd-dev libx265-209
libgtksourceview-3.0-1 openjdk-23-jre
libgtksourceview-3.0-common openjdk-23-jre-headless
libgtksourceviewmm-3.0-0v5 python3-appdirs
Veuillez utiliser « sudo apt autoremove » pour les supprimer.

Upgrading:
imagemagick libsmbclient0 python3-samba samba-common-bin
```

Création d'une structure de dossiers :

Création d'un dossier cybersec avec trois sous-dossiers : scan, logs, scripts :



```
lordson78@pentest: ~/Bureau/cybersec
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec]
$ git clone https://github.com/Leroyson007/cybersec.git
Clonage dans 'cybersec'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Réception d'objets: 100% (3/3), fait.

(lordson78@pentest)-[~/Bureau/cybersec]
$ mkdir scan

(lordson78@pentest)-[~/Bureau/cybersec]
$ mkdir logs

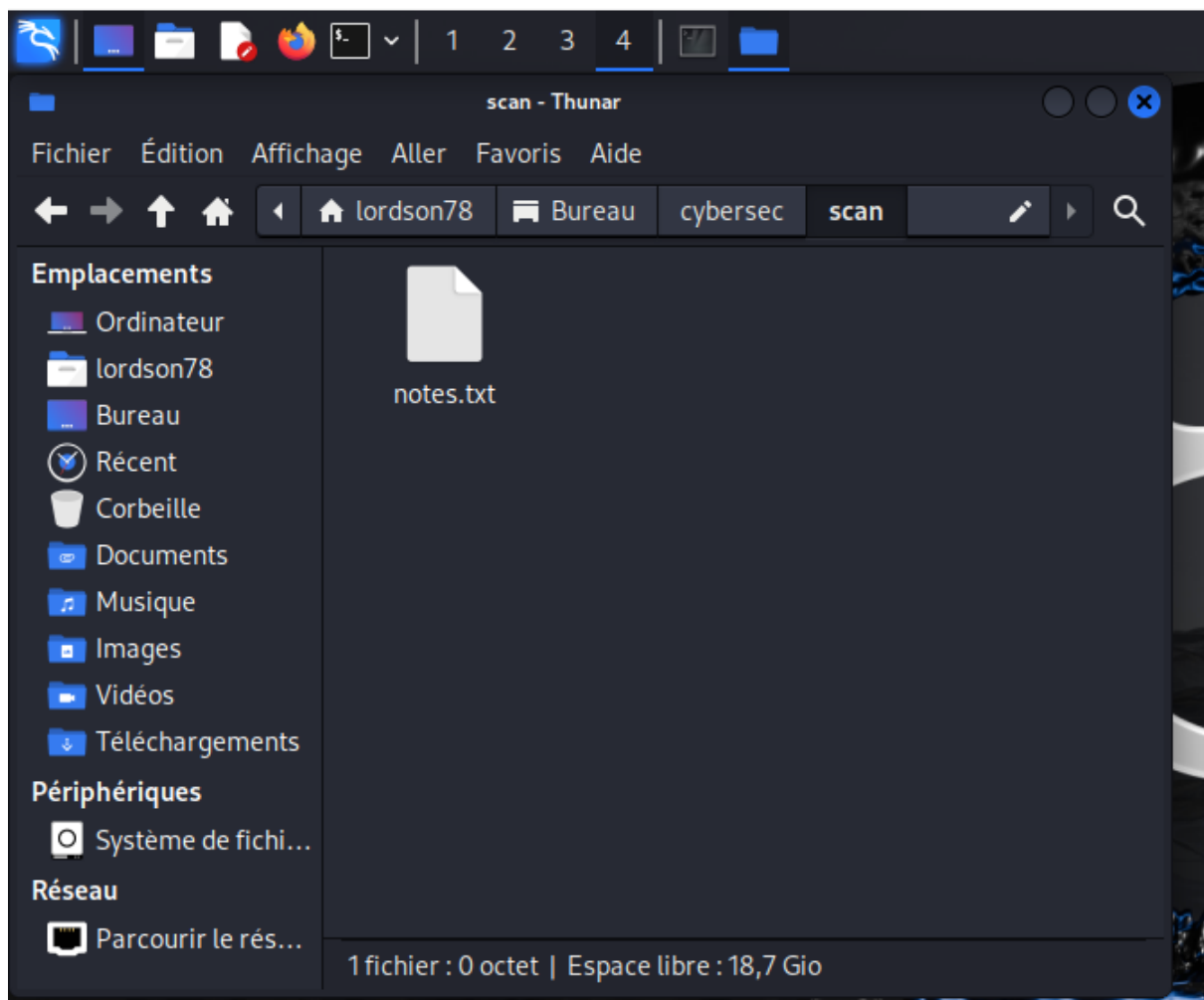
(lordson78@pentest)-[~/Bureau/cybersec]
$ mkdir scripts

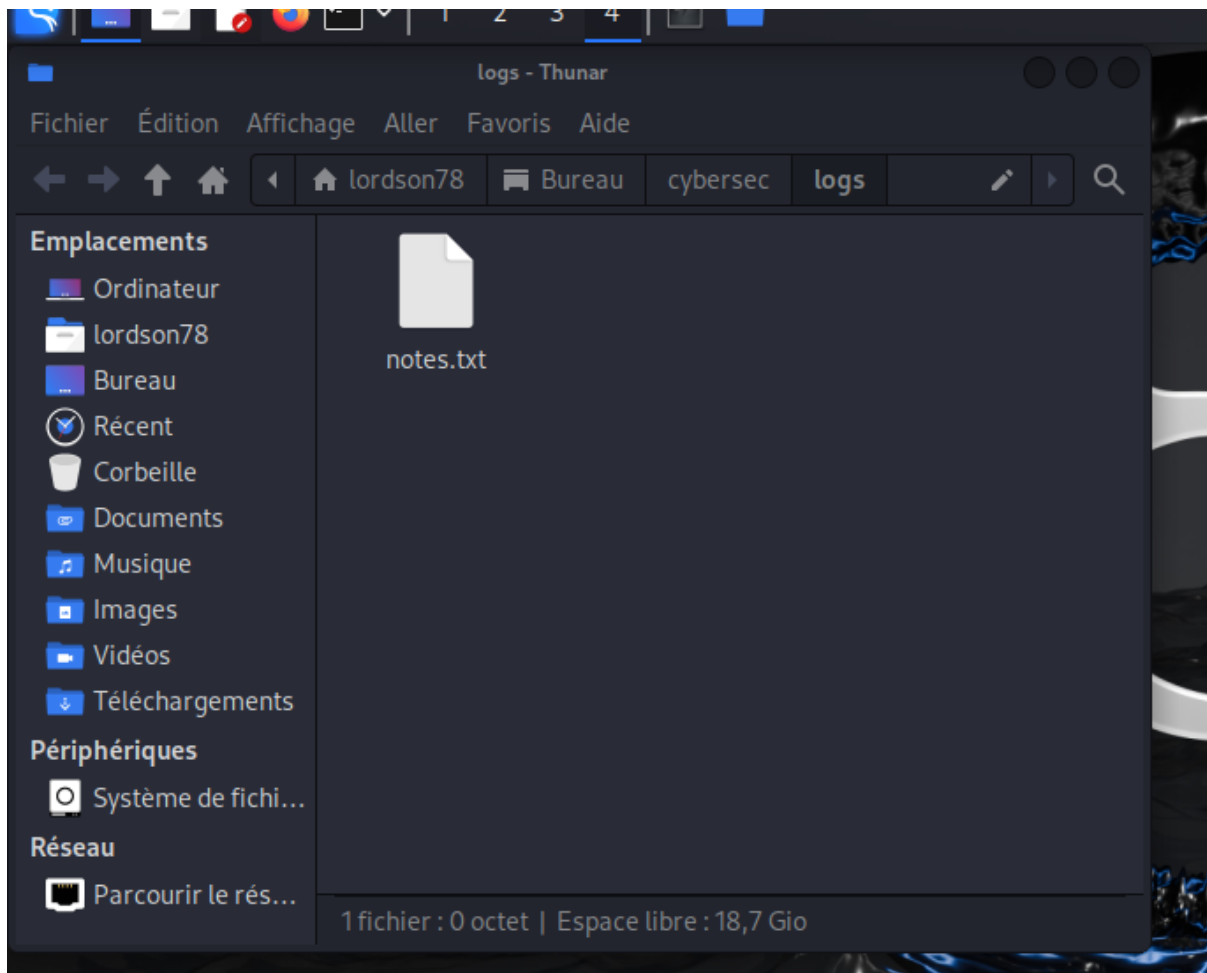
(lordson78@pentest)-[~/Bureau/cybersec]
$
```

Ajout d'un fichier notes.txt dans scan et logs :

```
lordson78@pentest: ~/Bureau/cybersec/scan
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$ touch notes.txt
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$
```

```
lordson78@pentest: ~/Bureau/cybersec/logs
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/logs]
$ touch notes.txt
(lordson78@pentest)-[~/Bureau/cybersec/logs]
$
```





Ajout du contenu dans les fichiers textes (notes.txt), puis affichez le contenu des fichiers :

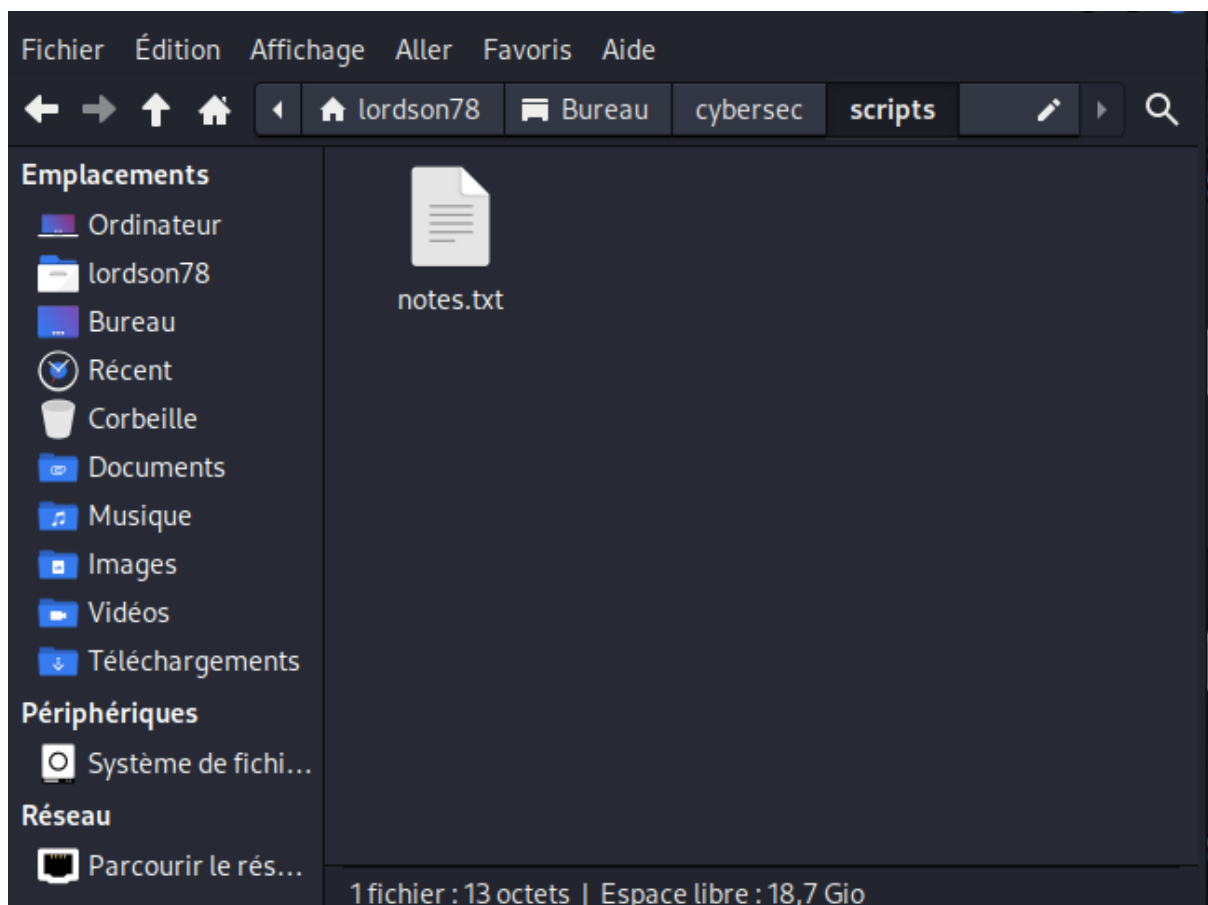
```
lordson78@pentest: ~/Bureau/cybersec/scan
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$ echo "Pour le bien">notes.txt
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$ cat notes.txt
Pour le bien
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$
```

```
lordson78@pentest: ~/Bureau/cybersec/logs
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/logs]
$ echo "Kali linux">notes.txt
(lordson78@pentest)-[~/Bureau/cybersec/logs]
$ cat notes.txt
Kali linux
(lordson78@pentest)-[~/Bureau/cybersec/logs]
$
```

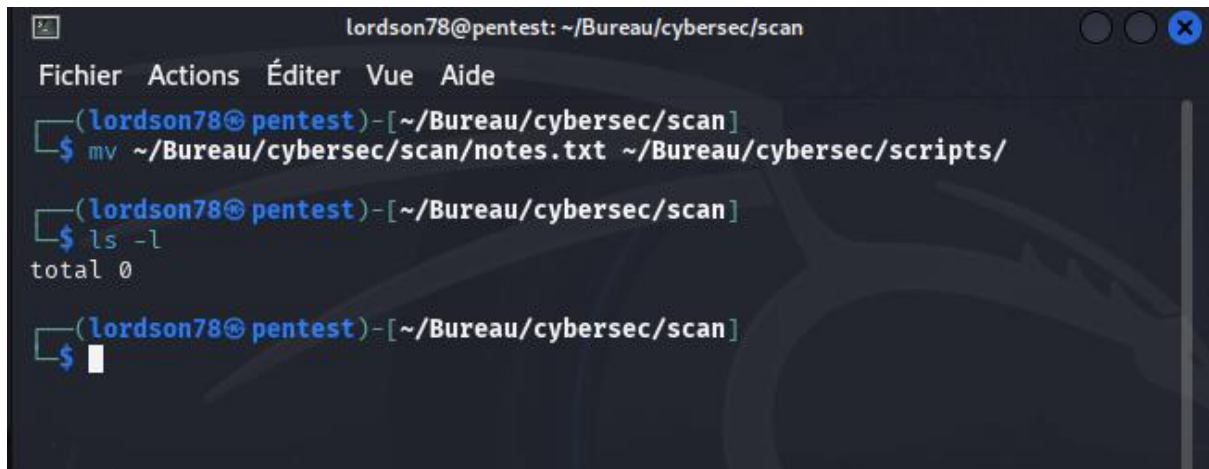
Copie du fichier (notes.txt) dans le sous-dossier scripts :

Vérification si le fichier a été copié :

```
(lordson78@pentest)-[~/Bureau/cybersec]
$ cp ~/Bureau/cybersec/scan/notes.txt ~/Bureau/cybersec/scripts/
```



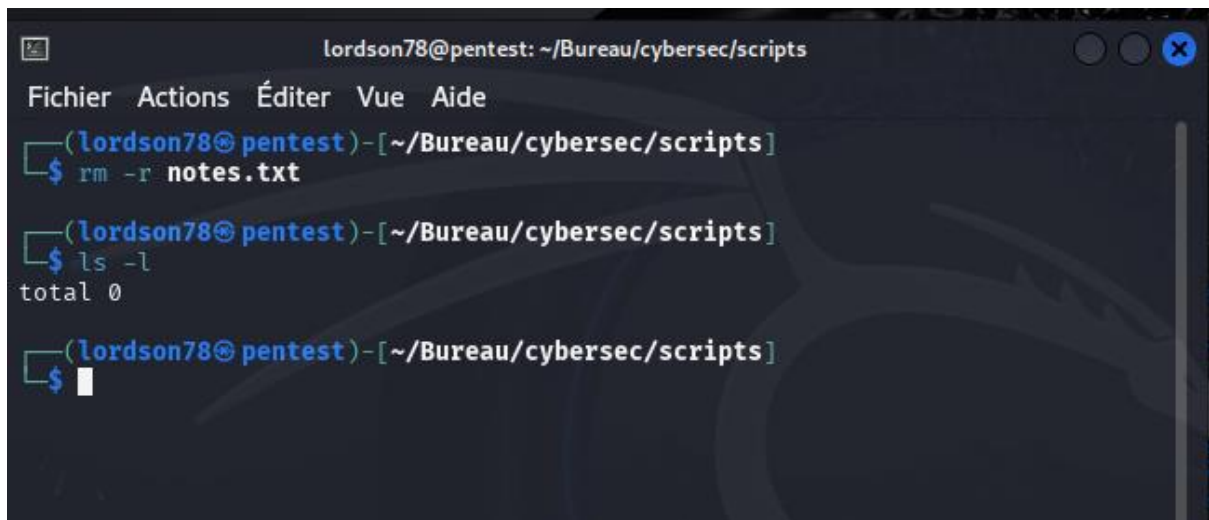
Déplacez le fichier (notes.txt) dans le sous-dossier scan

A terminal window titled 'lordson78@pentest: ~/Bureau/cybersec/scan'. It shows three commands: moving a file, listing the directory, and a prompt for the next command.

```
lordson78@pentest: ~/Bureau/cybersec/scan
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$ mv ~/Bureau/cybersec/scan/notes.txt ~/Bureau/cybersec/scripts/
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$ ls -l
total 0
(lordson78@pentest)-[~/Bureau/cybersec/scan]
$
```

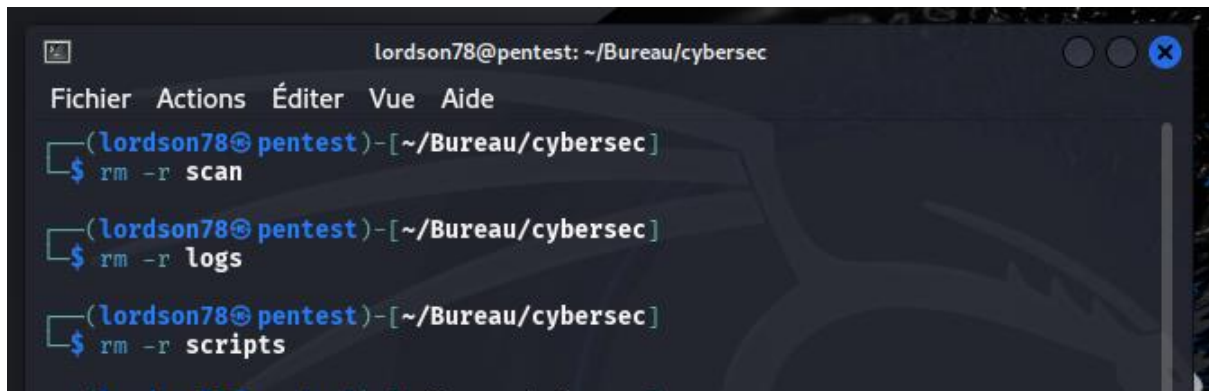
Supprimez le fichier (notes.txt) dans le sous-dossier scripts

Vérification si le fichiers a été supprimé :

A terminal window titled 'lordson78@pentest: ~/Bureau/cybersec/scripts'. It shows three commands: removing a file, listing the directory, and a prompt for the next command.

```
lordson78@pentest: ~/Bureau/cybersec/scripts
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec/scripts]
$ rm -r notes.txt
(lordson78@pentest)-[~/Bureau/cybersec/scripts]
$ ls -l
total 0
(lordson78@pentest)-[~/Bureau/cybersec/scripts]
$
```

Suppression des sous-dossiers : scan, logs, scripts :



```
lordson78@pentest: ~/Bureau/cybersec
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec]
$ rm -r scan
(lordson78@pentest)-[~/Bureau/cybersec]
$ rm -r logs
(lordson78@pentest)-[~/Bureau/cybersec]
$ rm -r scripts
```

Vérification si les sous-dossiers ont été supprimés :



```
(lordson78@pentest)-[~/Bureau/cybersec]
$ ls
cybersec
(lordson78@pentest)-[~/Bureau/cybersec]
$
```

Scanner un réseau :

ifconfig ou ip a : Affiche les informations réseau :

```
lordson78@pentest: ~/Bureau
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
  ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
  roup default qlen 1000
    link/ether 08:00:27:7c:6f:79 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82793sec preferred_lft 82793sec
    inet6 fe80::a00:27ff:fe7c:6f79/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(lordson78@pentest)-[~/Bureau]
$
```

Utilisez nmap pour scanner votre réseau local et identifier les appareils connectés :

```
lordson78@pentest: ~/Bureau
Fichier Actions Éditer Vue Aide
    valid_lft forever preferred_lft forever
(lordson78@pentest)-[~/Bureau]
$ nmap -sv 192.168.0.1
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3], ...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given po
  rts
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2], ...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
```

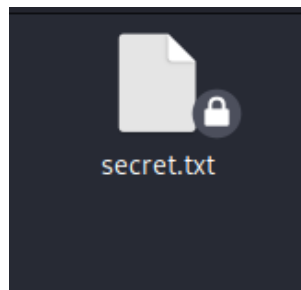

Manipuler les permissions :

Créez un fichier secret.txt et changez ses permissions pour qu'il ne soit accessible qu'en lecture par le propriétaire :

```
(lordson78@pentest)-[~/Bureau/cybersec]
$ touch secret.txt

(lordson78@pentest)-[~/Bureau/cybersec]
$ chmod 400 secret.txt

(lordson78@pentest)-[~/Bureau/cybersec]
$ ls -l
total 4
drwxrwxr-x 3 lordson78 lordson78 4096  7 fév 14:11 cybersec
-r----- 1 lordson78 lordson78    0  7 fév 14:55 secret.txt
```



Utiliser grep :

Créez un fichier log.txt avec des lignes de texte, puis utilisez grep pour rechercher un mot spécifique :

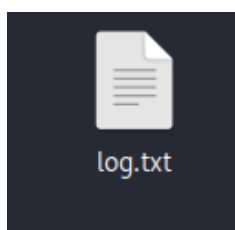
```
(lordson78@pentest)-[~/Bureau/cybersec]
$ touch log.txt

(lordson78@pentest)-[~/Bureau/cybersec]
$ echo "Nous sommes des etudiants">log.txt

(lordson78@pentest)-[~/Bureau/cybersec]
$ cat log.txt
Nous sommes des etudiants

(lordson78@pentest)-[~/Bureau/cybersec]
$ grep "etudiants" log.txt
Nous sommes des etudiants

(lordson78@pentest)-[~/Bureau/cybersec]
$
```



Exécution des commandes :

df -h

```
lordson78@pentest: ~/Bureau/cybersec
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec]
$ df -h
Sys. de fichiers Taille Utilisé Dispo Uti% Monté sur
udev                925M      0  925M   0% /dev
tmpfs               198M    1000K  197M   1% /run
/dev/sda1           38G      17G   19G  47% /
tmpfs               988M     4,0K  988M   1% /dev/shm
tmpfs                5,0M      0   5,0M   0% /run/lock
tmpfs                1,0M      0   1,0M   0% /run/credentials/systemd-journald.
service
tmpfs               988M     8,0K  988M   1% /tmp
tmpfs                1,0M      0   1,0M   0% /run/credentials/getty@tty1.servic
e
tmpfs               198M    116K  198M   1% /run/user/1000
```

du -sh

```
(lordson78@pentest)-[~/Bureau/cybersec]
$ du -sh
204K .
```

free -h

ps aux

```
Fichier Actions Éditer Vue Aide
(lordson78@pentest)-[~/Bureau/cybersec]
$ free -h
              total        utilisé        libre        partagé tamp/cache    dispon
ible
Mem:          1,9Gi         611Mi         1,0Gi          10Mi          455Mi         1,
3Gi
Échange:       2,0Gi           0B         2,0Gi

(lordson78@pentest)-[~/Bureau/cybersec]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  0.6  23176 14092 ?        Ss   13:51   0:01 /sbin/init
root           2  0.0  0.0      0      0 ?        S    13:51   0:00 [kthreadd]
root           3  0.0  0.0      0      0 ?        S    13:51   0:00 [pool_work
root           4  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/R
root           5  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/R
root           6  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/R
root           7  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/R
root           8  0.0  0.0      0      0 ?        I    13:51   0:00 [kworker/0
root          10  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/0
root          12  0.0  0.0      0      0 ?        I<   13:51   0:00 [kworker/R
root          13  0.0  0.0      0      0 ?        I    13:51   0:00 [rcu_tasks
root          14  0.0  0.0      0      0 ?        I    13:51   0:00 [rcu_tasks
root          15  0.0  0.0      0      0 ?        I    13:51   0:00 [rcu_tasks
root          16  0.0  0.0      0      0 ?        S    13:51   0:00 [ksoftirqd
root          17  0.0  0.0      0      0 ?        R    13:51   0:00 [rcu_preem
root          18  0.0  0.0      0      0 ?        S    13:51   0:00 [rcu_exp_p
root          19  0.0  0.0      0      0 ?        S    13:51   0:00 [rcu_exp_g
```

lspci

```
(lordson78@pentest)-[~/Bureau/cybersec]
$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:02.0 VGA compatible controller: VMware SVGA II Adapter
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Contr
oller (rev 02)
00:04.0 System peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Servi
ce
00:05.0 Multimedia audio controller: Intel Corporation 82801AA AC'97 Audio Co
ntroller (rev 01)
00:06.0 USB controller: Apple Inc. KeyLargo/Intrepid USB
00:07.0 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:0b.0 USB controller: Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family)
USB2 EHCI Controller
00:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA C
ontroller [AHCI mode] (rev 02)
```

```
sudo apt install traceroute
```

```

root@pentest: ~
Fichier Actions Éditer Vue Aide

(root@pentest)-[~] 0.0 2576 1724 ? Ss 13:54 0:00 fusermount
# sudo apt install traceroute 2 19708 ? Ssl 13:54 0:00 /usr/libe
traceroute est déjà la version la plus récente (1:2.1.6-1). 0:00 /usr/bin/
Les paquets suivants ont été installés automatiquement et ne sont plus nécess
aires :
imagemagick-6.q16 libhdf5-hl-100t64
libbfbio1 libjxl0.9
libc++1-19 libmagickcore-6.q16-7-extra
libc++abi1-19 libmagickcore-6.q16-7t64 [Natomia] (rev 02)
libcapstone4 libmagickwand-6.q16-7t64 [Natomia/Triton II]
libconfig++9v5 libmbdcrypto7t64 [EB/MB PIIX4 IDE (rev 01)
libconfig9 libpaper1 [svGA II Adapter
libdirectfb-1.7-7t64 libqt5*11extras5 [32540EM Gigabit Ethernet Cont
libegl-dev libsuperlu6
libfmt9 libtag1v5 [Gambertung GmbH VirtualBox Guest Servi
libgl1-mesa-dev libtag1v5-vanilla
libgles-dev libtagc0 [Intel Corporation 82801AA AC'97 Audio Co
libgles1 libunwind-19
libglvnd-core-dev libwebrtc-audio-processing1
libglvnd-dev libx265-209 [3/MB PIIX4 ACPI (rev 08)
libgtksourceview-3.0-1 openjdk-23-jre [1GB/1GB/FR/FW/FRW (ICH6 Family
libgtksourceview-3.0-common openjdk-23-jre-headless
libgtksourceviewmm-3.0-0v5 python3-appdirs [10MB/MBM (ICH8M/ICH8M-E) SATA C
Veuillez utiliser « sudo apt autoremove » pour les supprimer.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0

```

traceroute google.com

```
00:00:00.000000 0:0 0:0 2370 1724 0 0 13:54 0:00 /usr/sbin/ps aux
(root@pentest)-[~] 0.9 406992 19708 0 0 0 0 13:54 0:00 /usr/libexec/ps
# traceroute google.com: 10256 6340 prs/0 0 0 13:54 0:00 /usr/bin/z
traceroute to google.com (142.250.189.142), 30 hops max, 60 byte packets
 1 10.0.2.2 (10.0.2.2) 2.546 ms 3.180 ms 2.982 ms 13:57 0:00 ps aux
 2 * * *
 3 * * * n78@pentest)-[~/Bureau/cybersec]
 4 * * *
 5 * * * ast bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
 6 * * * A bridge: Intel Corporation 82371AB/EB/MB PIIX3 ISA [Natoma/Triton II]
 7 * * * DE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
 8 * * * A compatible controller: VMware SVGA II Adapter
 9 * * * hernet controller: Intel Corporation 82540EM Gigabit Ethernet Contr
10 * * * (rev 02)
11 * * * ystem peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Servi
12 * * *
13 * * * ultimedia audio controller: Intel Corporation 82801AA AC'97 Audio Co
14 * * * (rev 01)
15 * * * B controller: Apple Inc. KeyLargo/Intrepid USB
16 * * * ridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
17 * * * B controller: Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family)
18 * * * Controller
19 * * * ATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA C
20 * * * [AHCI mode] (rev 02)
21 * * *
22 * * * n78@pentest)-[~/Bureau/cybersec]
23 * * * pi install traceroute
24 * * * de passe de lordson78 : []
```

netstat -tuln

```
00:00:00.000000 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
(root@pentest)-[~] Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family)
# netstat -tuln
Connexions Internet actives (seulement serveurs)
Proto Recv-Q Send-Q Adresse locale Adresse distante Etat
```

ss -tuln

```
(root@pentest)-[~] Apple Inc. KeyLargo/Intrepid USB
# ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
USB2 EHCI Controller
```


journalctl

```
root@pentest: ~  
Fichier Actions Éditer Vue Aide  
root 2533 0.0 0.0 2576 1724 ? Ss 13:54 0:00 fusermount  
(root@pentest)-[~] 0.9 406992 19708 ? Ssl 13:54 0:00 /usr/libex  
# journalctl 0.3 0.3 10256 6340 pts/0 Ss 13:54 0:00 /usr/bin/j  
fév 07 10:34:51 pentest kernel: Linux version 6.11.2-amd64 (devel@kali.org) >  
fév 07 10:34:51 pentest kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.>  
fév 07 10:34:51 pentest kernel: [Firmware Bug]: TSC doesn't count with P0 fr>  
fév 07 10:34:51 pentest kernel: BIOS-provided physical RAM map:  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x0000000000009fc00-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x000000000007ffff000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x0000000000fec00000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x0000000000fee00000-0x00000000>  
fév 07 10:34:51 pentest kernel: BIOS-e820: [mem 0x0000000000fffc0000-0x00000000>  
fév 07 10:34:51 pentest kernel: NX (Execute Disable) protection: active  
fév 07 10:34:51 pentest kernel: APIC: Static calls initialized AC'97 Audio Co  
fév 07 10:34:51 pentest kernel: SMBIOS 2.5 present.  
fév 07 10:34:51 pentest kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIO>  
fév 07 10:34:51 pentest kernel: DMI: Memory slots populated: 0/0  
fév 07 10:34:51 pentest kernel: Hypervisor detected: KVM  
fév 07 10:34:51 pentest kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00  
fév 07 10:34:51 pentest kernel: kvm-clock: using sched offset of 11046626128>  
fév 07 10:34:51 pentest kernel: clocksource: kvm-clock: mask: 0xffffffffffff>  
fév 07 10:34:51 pentest kernel: tsc: Detected 2096.060 MHz processor  
fév 07 10:34:51 pentest kernel: e820: update [mem 0x000000000-0x00000fff] usa>  
fév 07 10:34:51 pentest kernel: e820: remove [mem 0x000a00000-0x000fffff] usa>  
fév 07 10:34:51 pentest kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
```

journalctl -f

```
(root@pentest)-[~]  
# journalctl -f ~ /Bureau/cybersec  
fév 10 14:06:57 pentest systemd[1]: Finished systemd-tmpfiles-clean.service -  
Cleanup of Temporary Directories.  
fév 10 14:09:01 pentest CRON[9799]: pam_unix(cron:session): session opened fo  
r user root(uid=0) by root(uid=0)  
fév 10 14:09:01 pentest CRON[9801]: (root) CMD ( [ -x /usr/lib/php/sessioncl  
ean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)  
fév 10 14:09:01 pentest CRON[9799]: pam_unix(cron:session): session closed fo  
r user root  
fév 10 14:09:10 pentest systemd[1]: Starting phpsessionclean.service - Clean  
php session files ...  
fév 10 14:09:10 pentest systemd[1]: phpsessionclean.service: Deactivated succ  
essfully.  
fév 10 14:09:10 pentest systemd[1]: Finished phpsessionclean.service - Clean  
php session files.  
fév 10 14:11:52 pentest systemd[1]: Starting apt-daily.service - Daily apt do  
wnload activities ...  
fév 10 14:11:52 pentest systemd[1]: apt-daily.service: Deactivated successf  
ully.  
fév 10 14:11:52 pentest systemd[1]: Finished apt-daily.service - Daily apt do  
wnload activities.
```

journalctl -b

```
root@pentest: ~  
Fichier Actions Éditer Vue Aide  
root 2533 0.0 0.0 2576 1724 ? Ss 13:54 0:00 fusermount  
(root@pentest)-[~]  
# journalctl -b 0.2 0.3 10256 6340 pts/0 Ss 13:54 0:00 /usr/bin/s  
fév 10 13:51:52 pentest kernel: Linux version 6.11.2-amd64 (devel@kali.org) >  
fév 10 13:51:52 pentest kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.>  
fév 10 13:51:52 pentest kernel: [Firmware Bug]: TSC doesn't count with P0 fr>  
fév 10 13:51:52 pentest kernel: BIOS-provided physical RAM map:  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x000000000000f0000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x00000000000100000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x0000000007fff0000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000>  
fév 10 13:51:52 pentest kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000>  
fév 10 13:51:52 pentest kernel: NX (Execute Disable) protection: active  
fév 10 13:51:52 pentest kernel: APIC: Static calls initialized AC'97 Audio Co  
fév 10 13:51:52 pentest kernel: SMBIOS 2.5 present.  
fév 10 13:51:52 pentest kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIO>  
fév 10 13:51:52 pentest kernel: DMI: Memory slots populated: 0/0 (8)  
fév 10 13:51:52 pentest kernel: Hypervisor detected: KVM (V/F/W (ICH6 Family))  
fév 10 13:51:52 pentest kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00  
fév 10 13:51:52 pentest kernel: kvm-clock: using sched offset of 7704445083 >  
fév 10 13:51:52 pentest kernel: clocksource: kvm-clock: mask: 0xffffffffffff>  
fév 10 13:51:52 pentest kernel: tsc: Detected 2096.060 MHz processor  
fév 10 13:51:52 pentest kernel: e820: update [mem 0x00000000-0x00000fff] usa>  
fév 10 13:51:52 pentest kernel: e820: remove [mem 0x000a0000-0x000fffff] usa>  
fév 10 13:51:52 pentest kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
```

journalctl -n 10

```
(root@pentest)-[~]  
# journalctl -n 10 audio controller: Intel Corporation 82801AA AC'97 Audio Co  
fév 10 14:09:01 pentest CRON[9799]: pam_unix(cron:session): session closed f>  
fév 10 14:09:10 pentest systemd[1]: Starting phpsessionclean.service - Clean>  
fév 10 14:09:10 pentest systemd[1]: phpsessionclean.service: Deactivated suc>  
fév 10 14:09:10 pentest systemd[1]: Finished phpsessionclean.service - Clean>  
fév 10 14:11:52 pentest systemd[1]: Starting apt-daily.service - Daily apt d>  
fév 10 14:11:52 pentest systemd[1]: apt-daily.service: Deactivated successfu>  
fév 10 14:11:52 pentest systemd[1]: Finished apt-daily.service - Daily apt d>  
fév 10 14:15:01 pentest CRON[12848]: pam_unix(cron:session): session opened >  
fév 10 14:15:01 pentest CRON[12850]: (root) CMD (command -v debian-sa1 > /de>  
fév 10 14:15:01 pentest CRON[12848]: pam_unix(cron:session): session closed >
```

date

```
(root@pentest)-[~]: Intel Co  
# date [AHCI mode] (rev 02)  
lun 10 fév 2025 14:16:11 EST  
root@pentest: ~/Bureau
```

timedatectl

```
(root@pentest)-[~]: controller: Intel Corporation 82801AA AC'97 Audio Co  
# timedatectl  
00:00:00 USB cont Local time: lun 2025-02-10 14:16:42 EST  
00:00:00 ACPI Universal time: lun 2025-02-10 19:16:42 UTC ACPI (rev 06)  
00:00:00 USB cont RTC time: lun 2025-02-10 19:16:41 PM/PM/FW/FW (ICH6 Family)  
00:00:00 USB cont Time zone: America/Port-au-Prince (EST, -0500)  
System clock synchronized: no  
RTC in local TZ: no  
NTP service: inactive  
root@pentest: ~/Bureau/cybersec  
(root@pentest)-[~]:
```

hostnamectl

```
~/Bureau/cybersec  
(root@pentest)-[~]  
# hostnamectl  
Static hostname: pentest  
Icon name: computer-vm  
Chassis: vm  
Machine ID: 6b1eda80ef364ac0a46f397575479025  
Boot ID: 3dc1de6e4d9e4de3a8afe96b5647667f  
Product UUID: 5c142dcc-c013-6d4f-919b-8101a98b253f  
Virtualization: oracle  
Operating System: Kali GNU/Linux Rolling  
Kernel: Linux 6.11.2-amd64  
Architecture: x86-64  
Hardware Vendor: innotek GmbH  
Hardware Model: VirtualBox  
Hardware Serial: 0  
Firmware Version: VirtualBox  
Firmware Date: Fri 2006-12-01  
Firmware Age: 18y 2month 1w 4d  
root@pentest: ~/Bureau/cybersec  
(root@pentest)-[~]:
```

